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c. Only some of the
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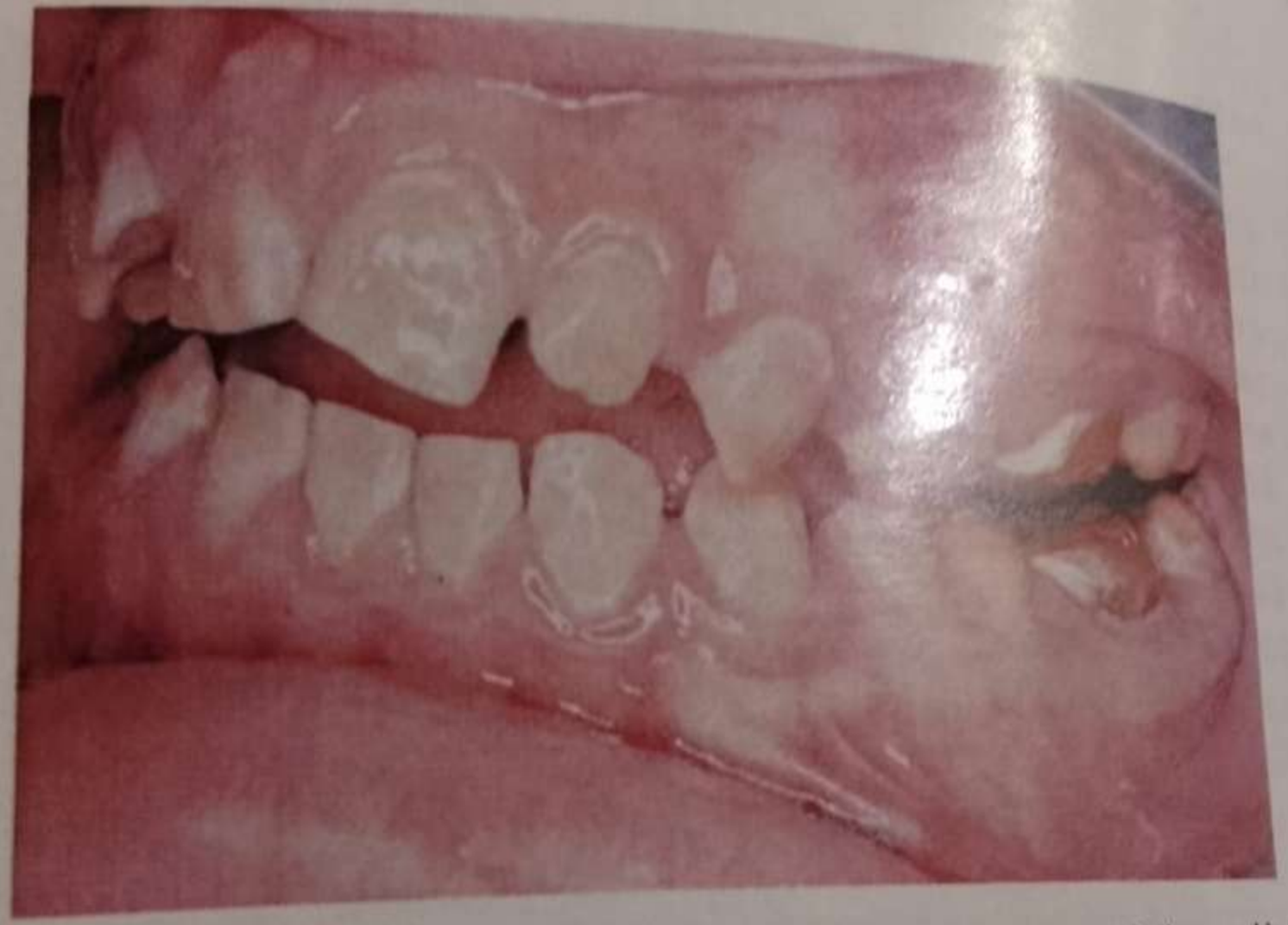
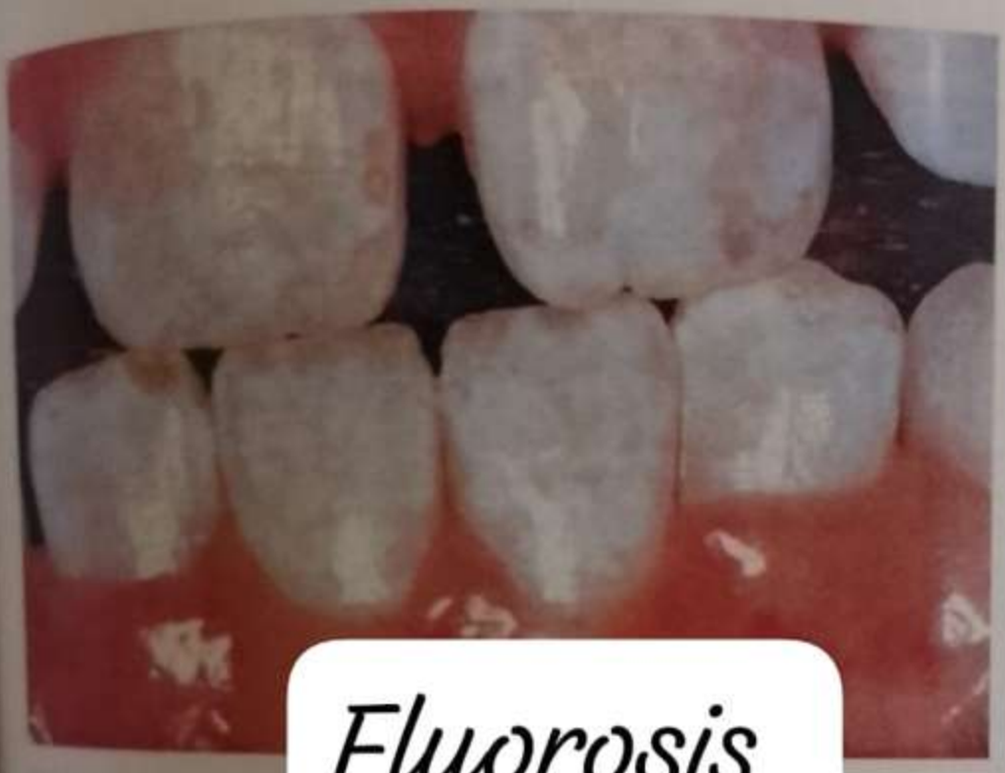


Fig. 22.32 Teeth from a patient with amelogenesis imperfecta. In this example the enamel is of apparently normal appearance in many of the teeth but seriously abnormal in the first molars, where much of the enamel has chipped off exposing the underlying dentine. The underlying dentine is usually normal and restorative procedures, especially crowns, can be used to maintain the dentition. Courtesy Dr M Ignelzi.



Fluorosis

Fig. 22.34 Teeth demonstrating … in much of the enamel becoming opaque in patches, giv…



Tetracycline

Fig. 22.35 Teeth stained as a result of tetracycline adm… during the entire enamel (and dentine) has become p… Tooth bleaching procedures do not usually greatly im… considerably, veneers, will do so. Courtesy Dr M Ignelzi.

of variable extent. In severe cases, the fluoritic areas may be brown and pitted (Fig. 22.34). The principal cha… of varying extent … hypomineralisation



Fig. 22.36 … incompatibili… is now a rare …

Lear…

Concep…

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3. unc… for…
4. kno… org…
5. unc…
6. des…
7. hav… for…

be …

Furthe…

Aoba, T. 19…



Fig. 22.36 Teeth showing haemoglobin pigmentation. Haemoglobin residues released during Rhesus factor incompatibility are taken up by the developing teeth, resulting in a brown stain. Because of effective screening this is now a rare occurrence. Courtesy Dr M Ignelzi.

Learning objectives

Concerning amelogenesis, the student should be able to:

1. appreciate the importance of epithelial–mesenchymal interactions
2. relate the changes in morphology during the ameloblast life cycle to its changing function
3. understand the significance of the Tomes process in terms of prism formation
4. know the composition of young enamel, particularly in terms of the organic matrix
5. understand the changes that take place during enamel maturation
6. [illegible] mental nature of amelogenesis
7. [illegible] that can occur during enamel [illegible]ically [illegible]togenesis.

The tissues that support the teeth, known collectively as the periodontium (Figs 7.1, 7.2), include the alveolar bone forming the tooth sockets, the periodontal ligament (a connective tissue that attaches the tooth to the alveolar bone) and the gingivae (the component of the oral mucosa that forms a collar around the tooth).

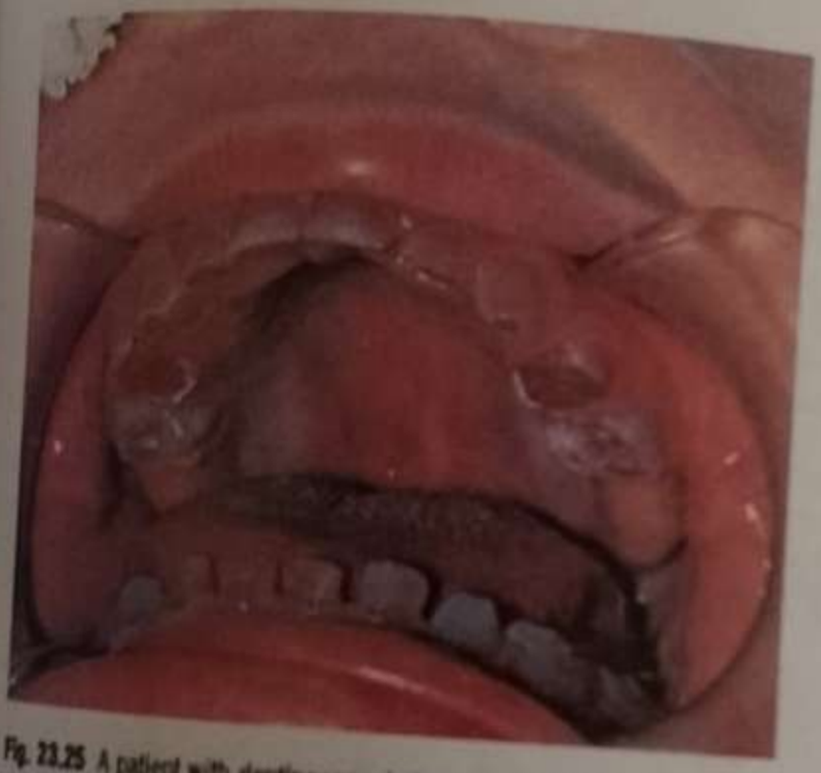
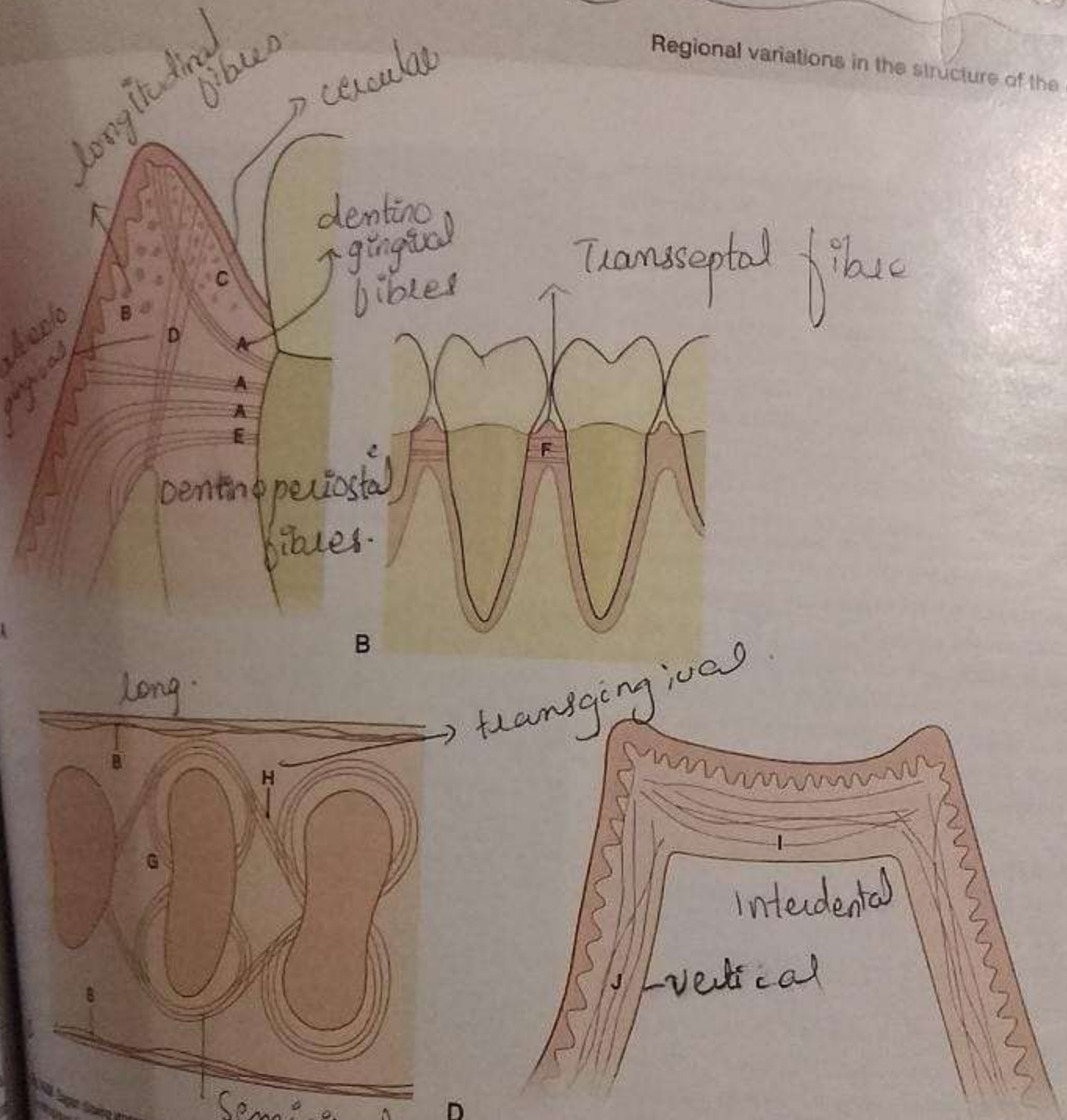


Fig. 23.25 A patient with dentinogenesis imperfecta. The teeth are malformed and are dark because of the absence of pulp chambers and the consequent loss of translucency. The enamel is normal but flakes off as it is only weakly attached to the dentine. In this example the enamel from the first molars has been lost. Courtesy M Ignelzi.

necrosis. Protecting the pulp and encouraging it to recover after caries has been removed is a principal aim of restorative dentistry. Tertiary dentine is the tissue that is laid down in response to a stimulus of any kind. It is a response rather than an age change. Tertiary dentine takes one of two forms depending on the severity of the stimulus. If the stimulus is mild and the original odontoblasts remain alive they will lay down a tubular form of tertiary dentine, *reactionary dentine*. If the stimulus is more severe and sufficient to kill the original odontoblasts, new odontoblasts will differentiate from pulpal stem cells and lay down another form of tertiary dentine, *reparative dentine*, which is atubular and bonelike.

One important difference between [illegible] new odontoblasts and those involved [illegible]

gingival
Interdental Tissue fibres



Semicircular

B

Sagittal sections showing arrangement of principal collagen fibre groups in the lamina propria of the gingiva. (A) buccolingual section, (B) mesiodistal section, (C) horizontal section, (D) buccolingual section along in
connective tissue; B = longitudinal fibres; C = circular fibres; D = alveologingival fibres; E = dentoperiosteal fibres; F = transseptal fibres; G = semicircular fibres; H = transgingival fibres; I = interdental fibres; J =

OSPE



Fig. 23.7 SEM showing the surface of predentine. The collagen fibrils (brown) form an interlacing network perpendicular to the odontoblast process (blue) (×4,000). Courtesy Prof. L Fonzi.





Min

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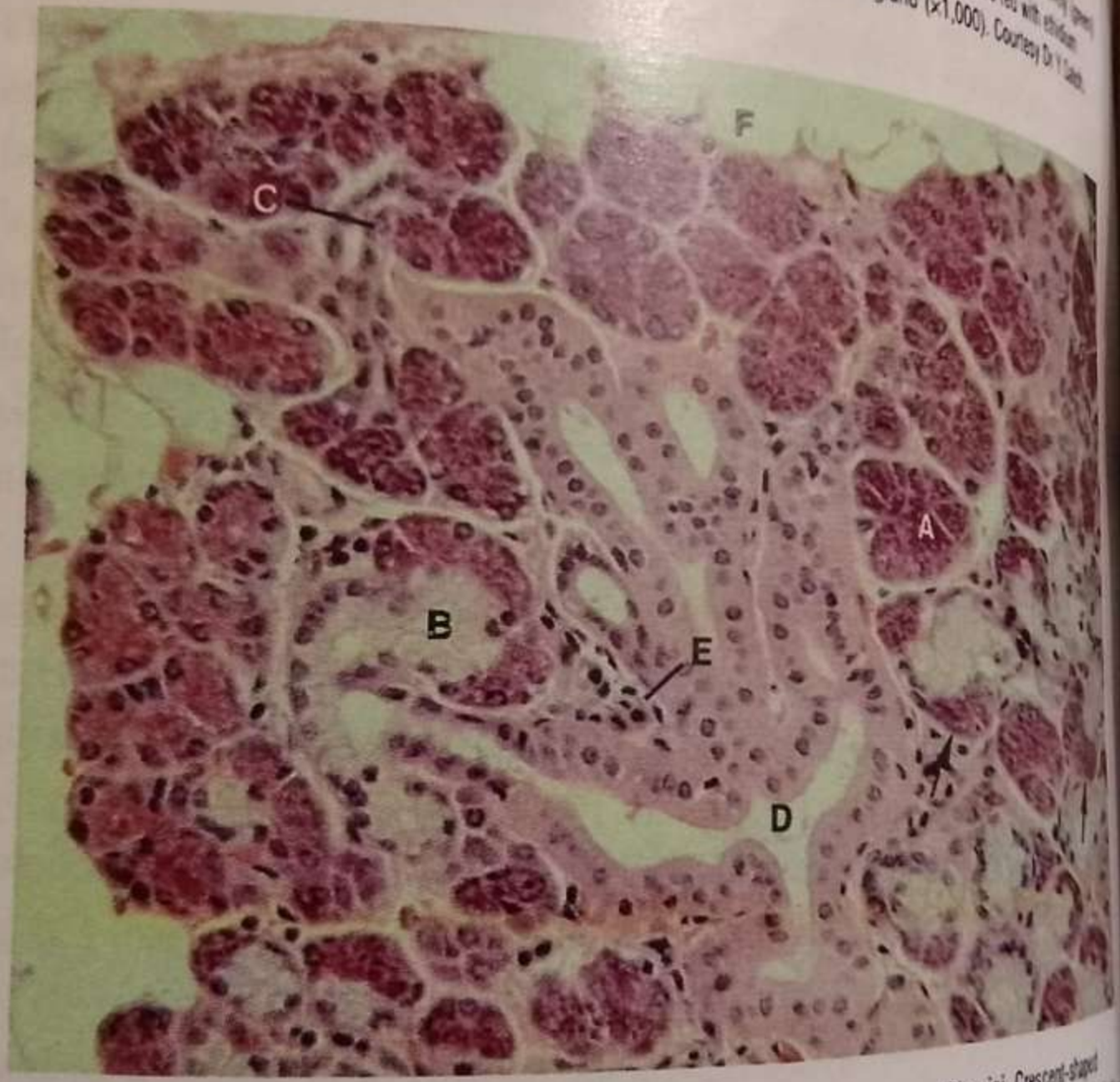


Fig. 16.26 Submandibular gland showing darker serous (A) and lighter mucous (B) acini. Crescent-shaped (but see text). C = intercalated duct; D = striated duct

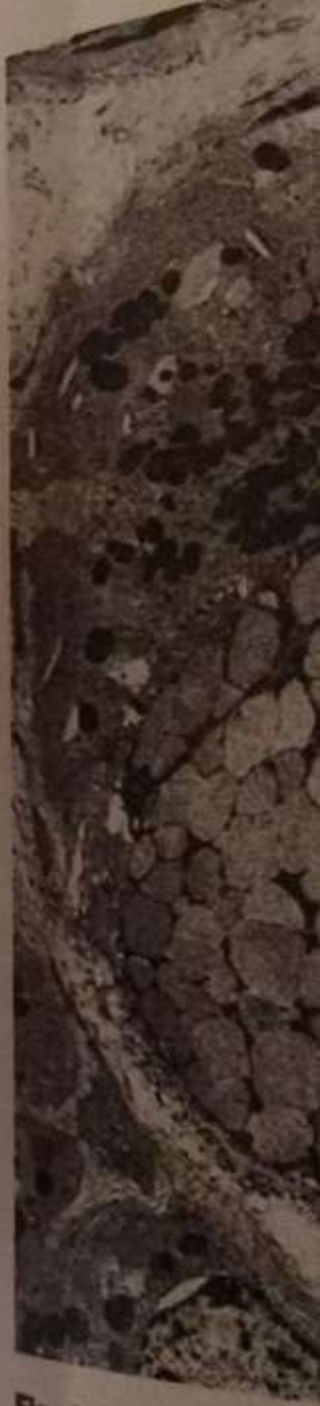
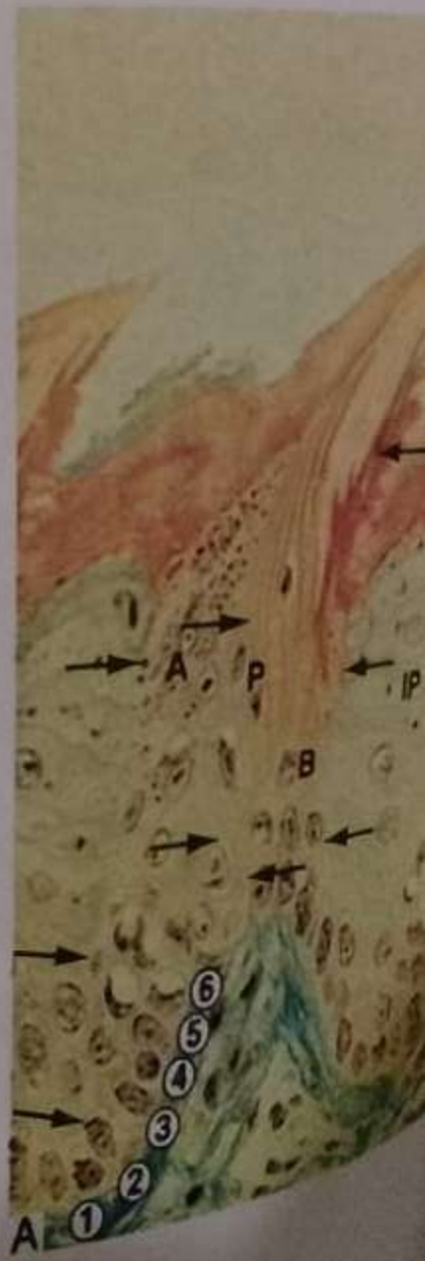
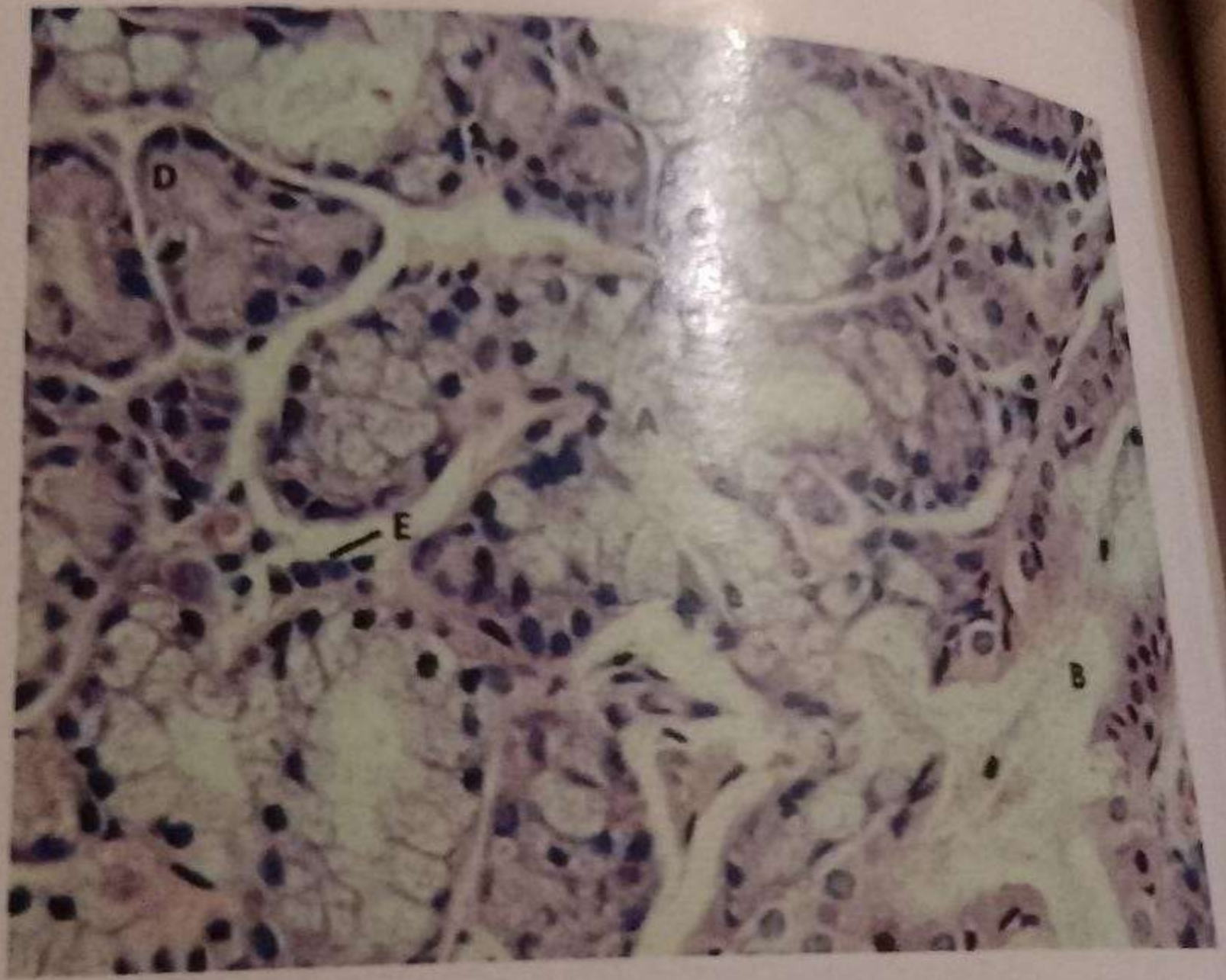


Fig. 16.27 TEM

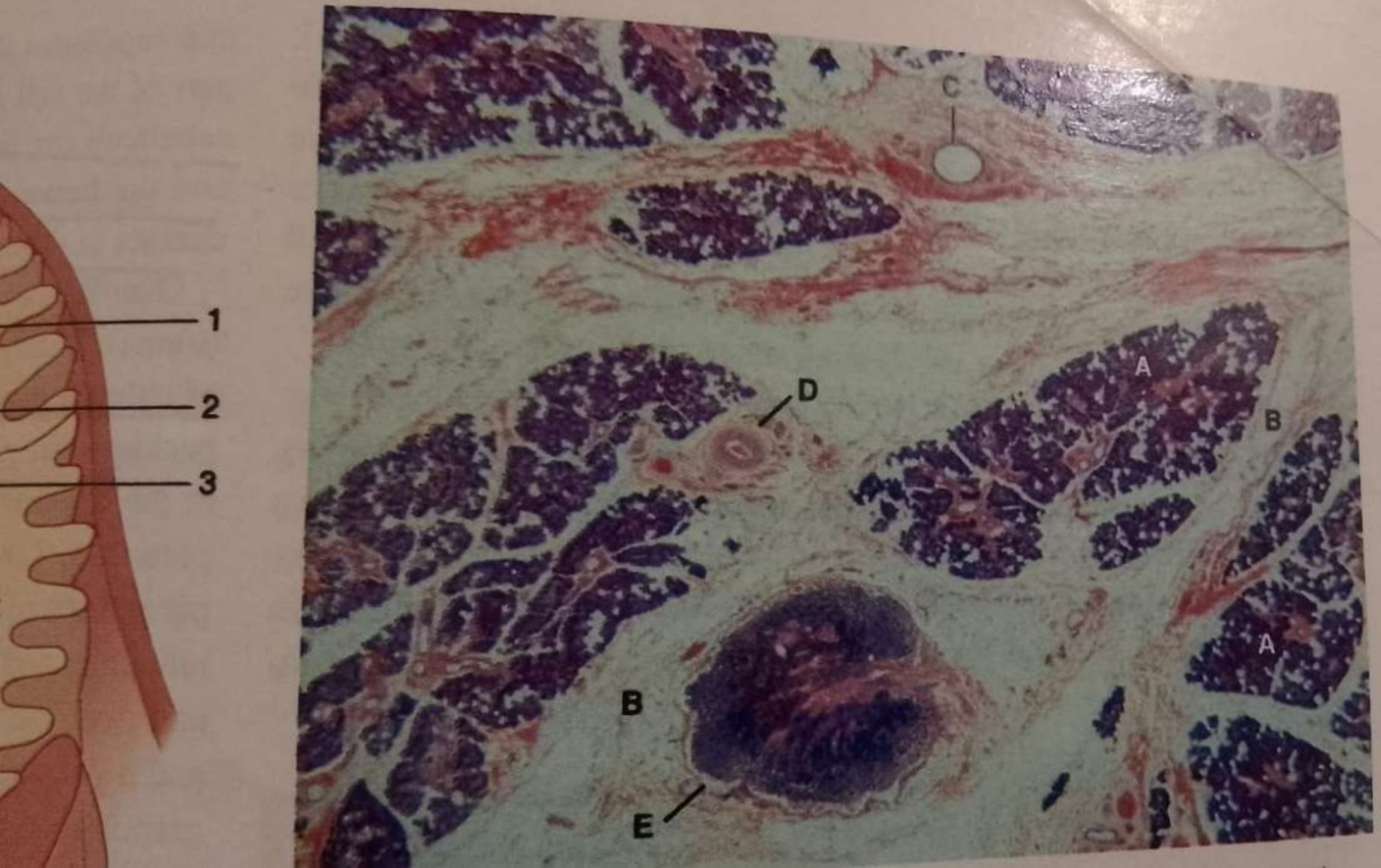


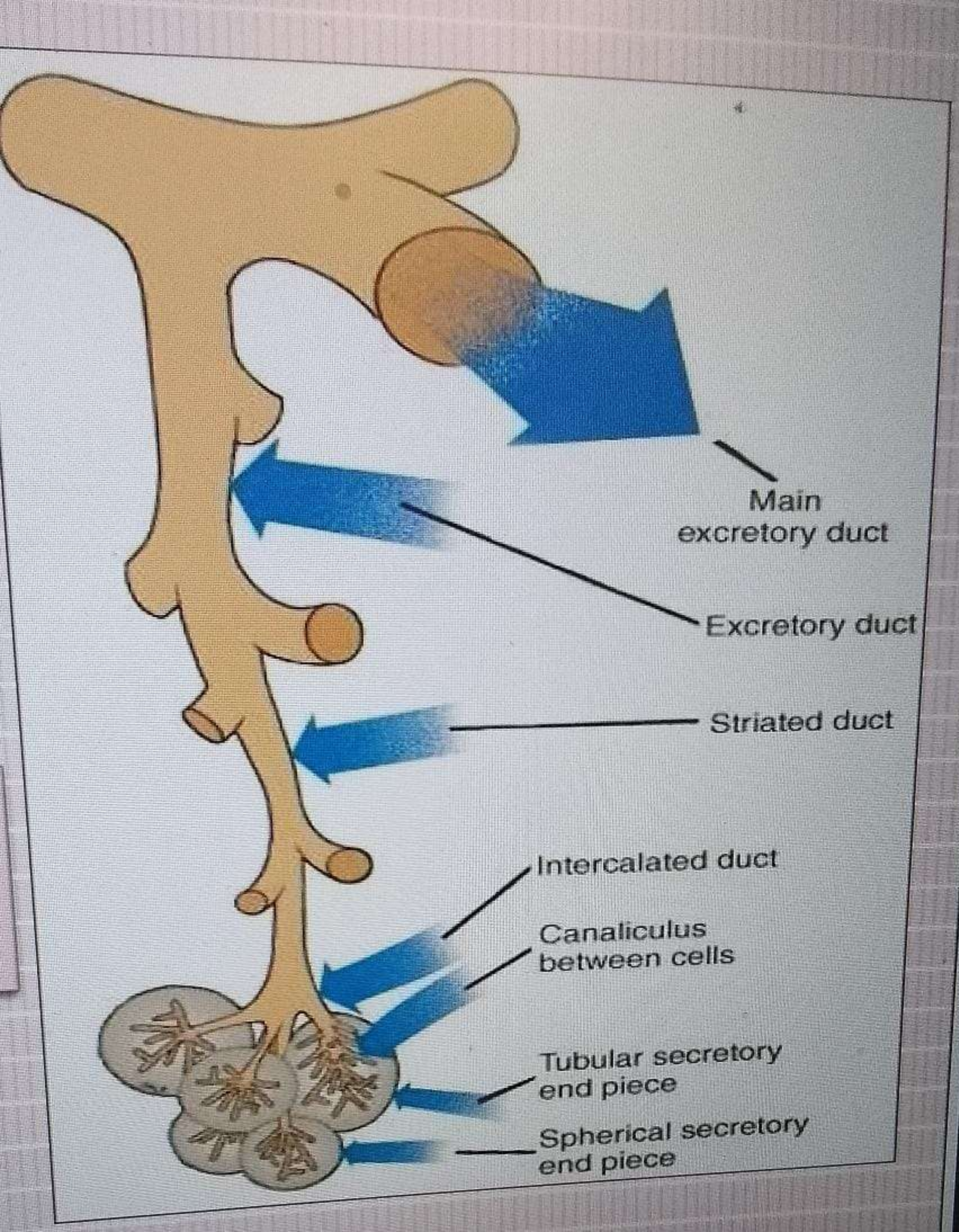
Fig. 14.66 Section showing dorsum of anterior two-thirds of tongue covered by keratinised filiform papillae (A) with nonkeratinised regions between (B) (H & E; ×35). Courtesy Prof. P R Morgan.





Pale-staining mucous acini (A) of sublingual gland drain into a collecting duct (B). The aqueous


1
2
3
C
D
A
B
A
B
E


Main
excretory duct
Excretory duct
Striated duct
Intercalated duct
Canaliculus
between cells
Tubular secretory
end piece
Spherical secretory
end piece



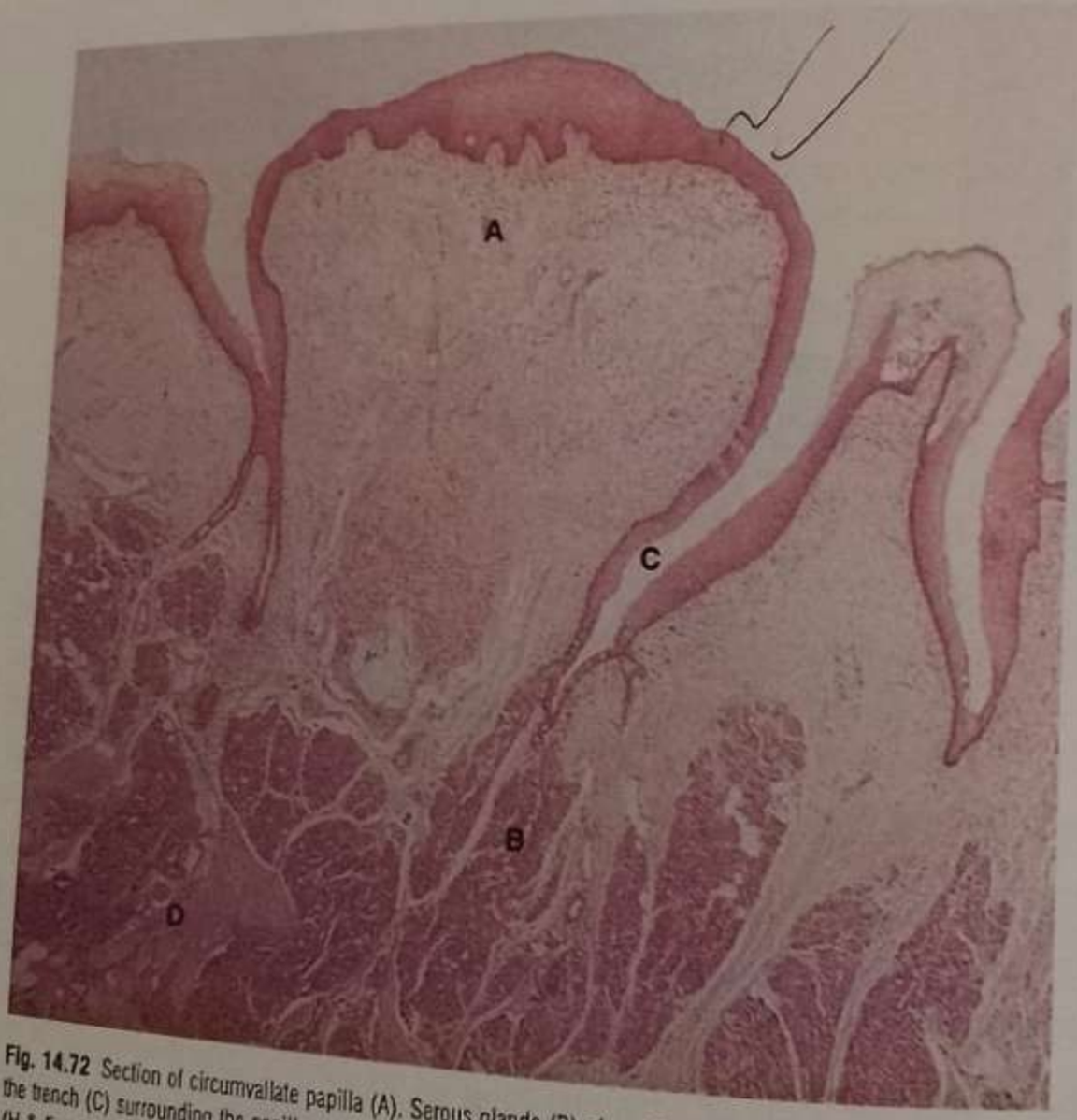


Fig. 14.72 Section of circumvallate papilla (A). Serous glands (B) of von Ebner empty via ducts into the base of the trench (C) surrounding the papilla, which is not raised above the surface of the tongue. D = muscle of tongue (H & E, ×35). Courtesy Prof. P R Morgan.

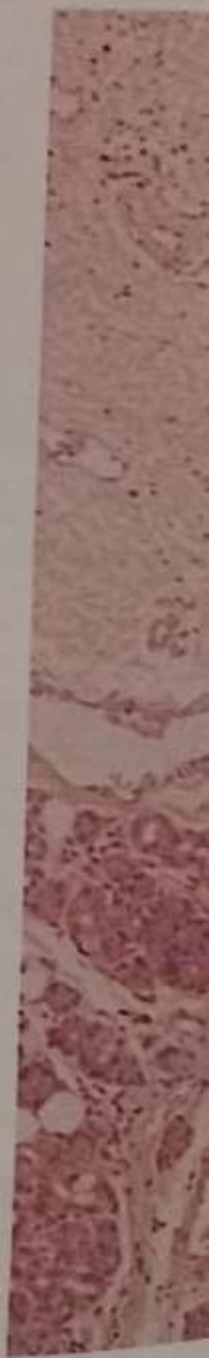


Fig. 14.73 Section o
base of the trench (C)

Clinical considerations

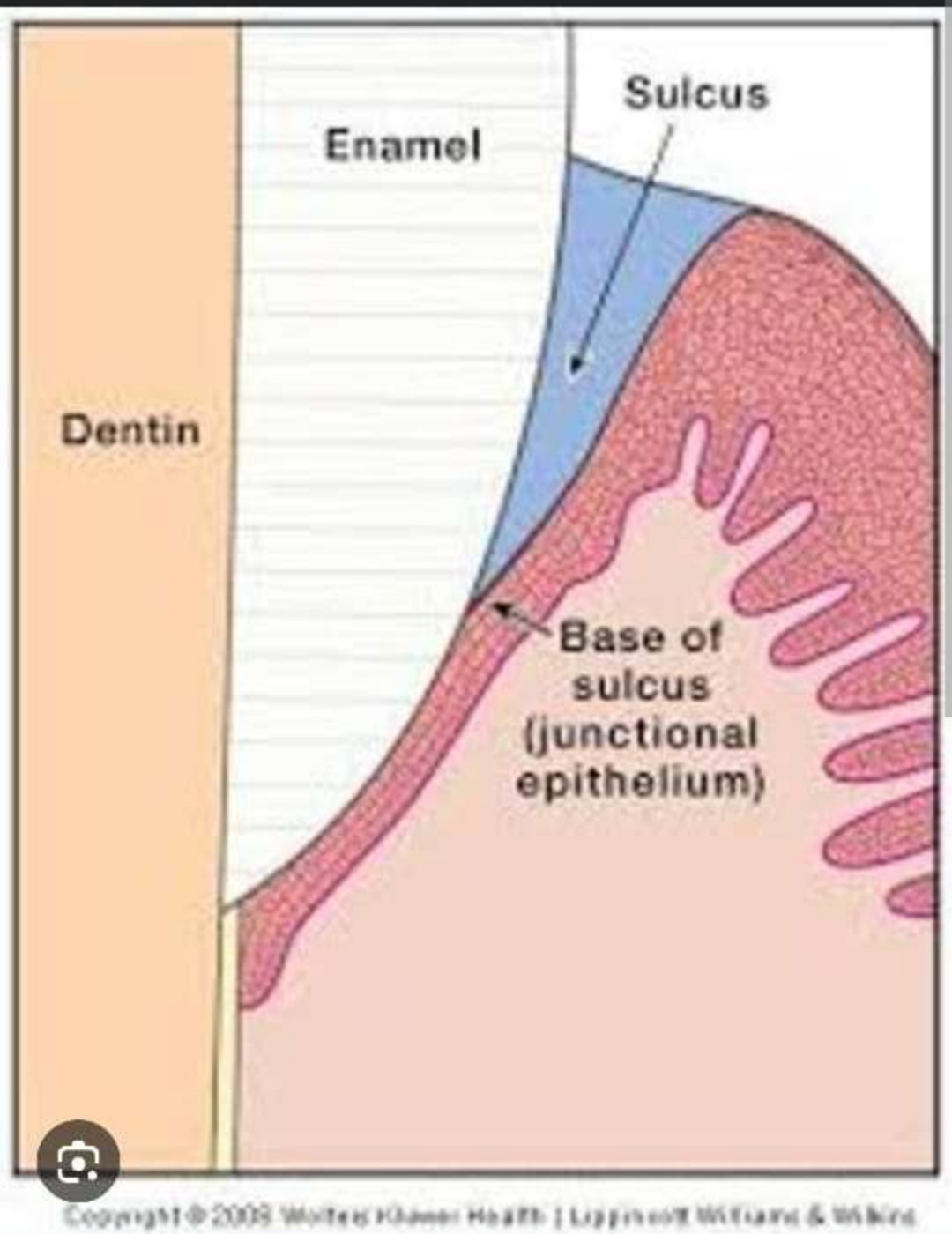
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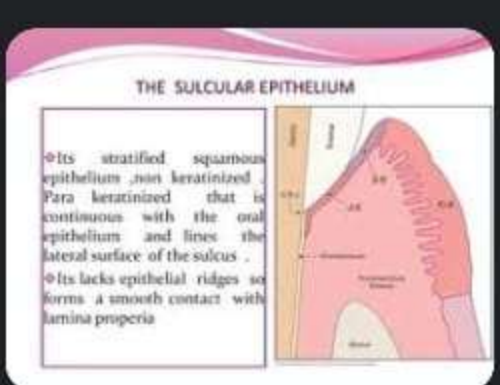
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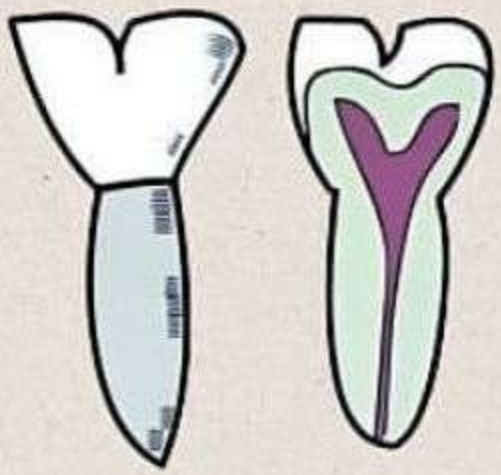




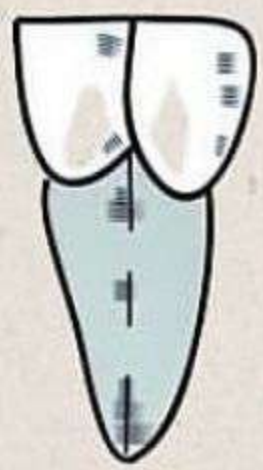
DEVELOPMENTAL ANOMALIES



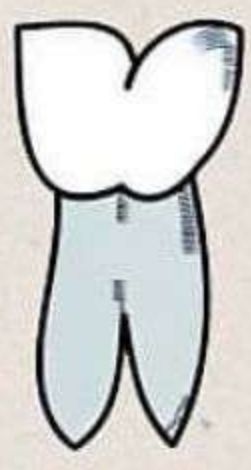
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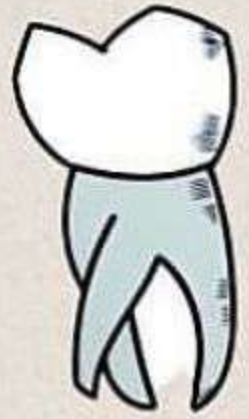
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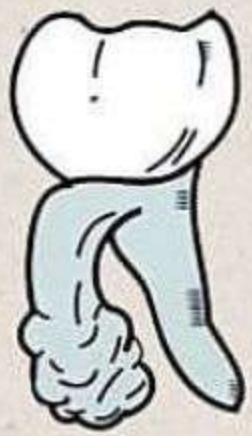
TAURODONTISM



DILACERATION

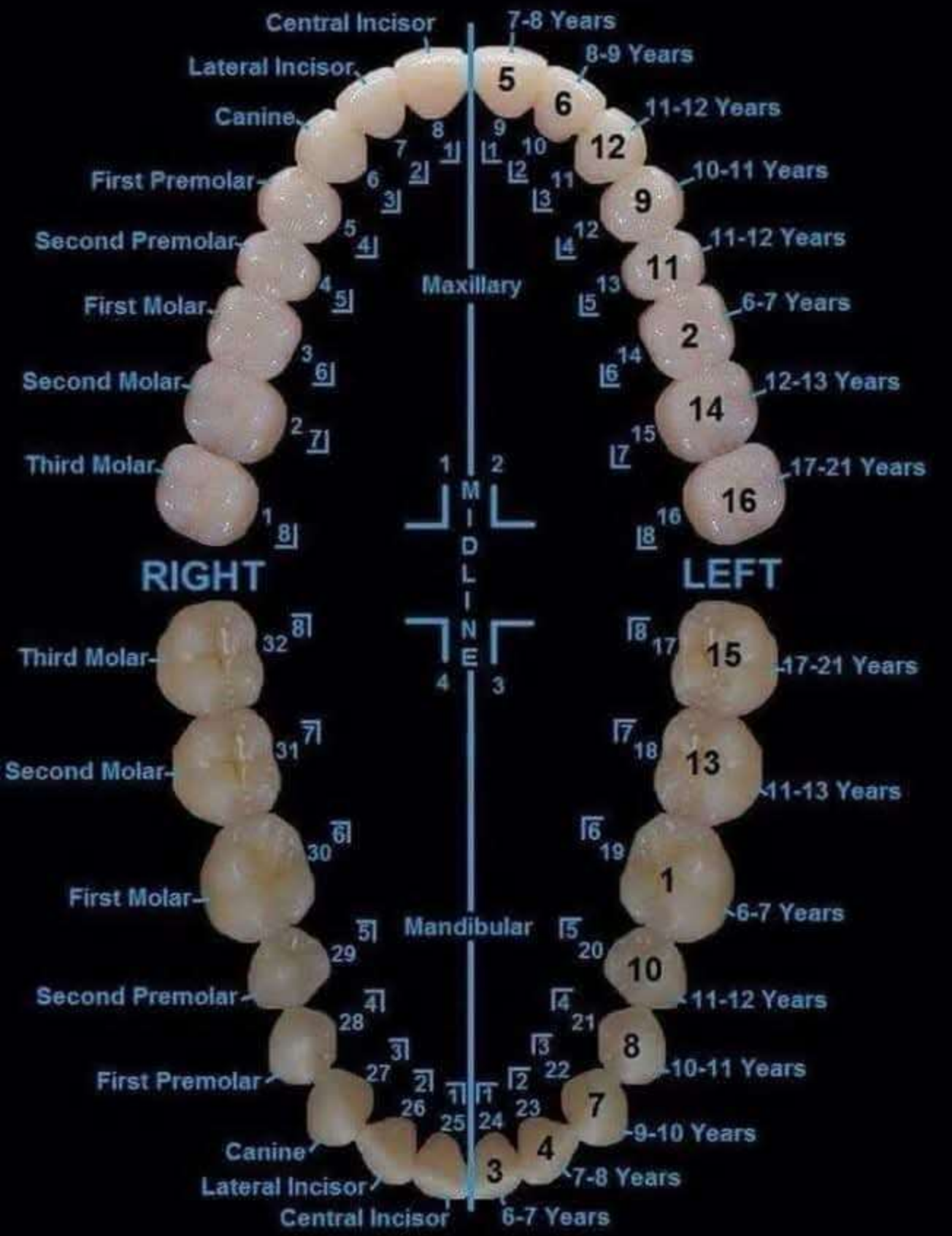


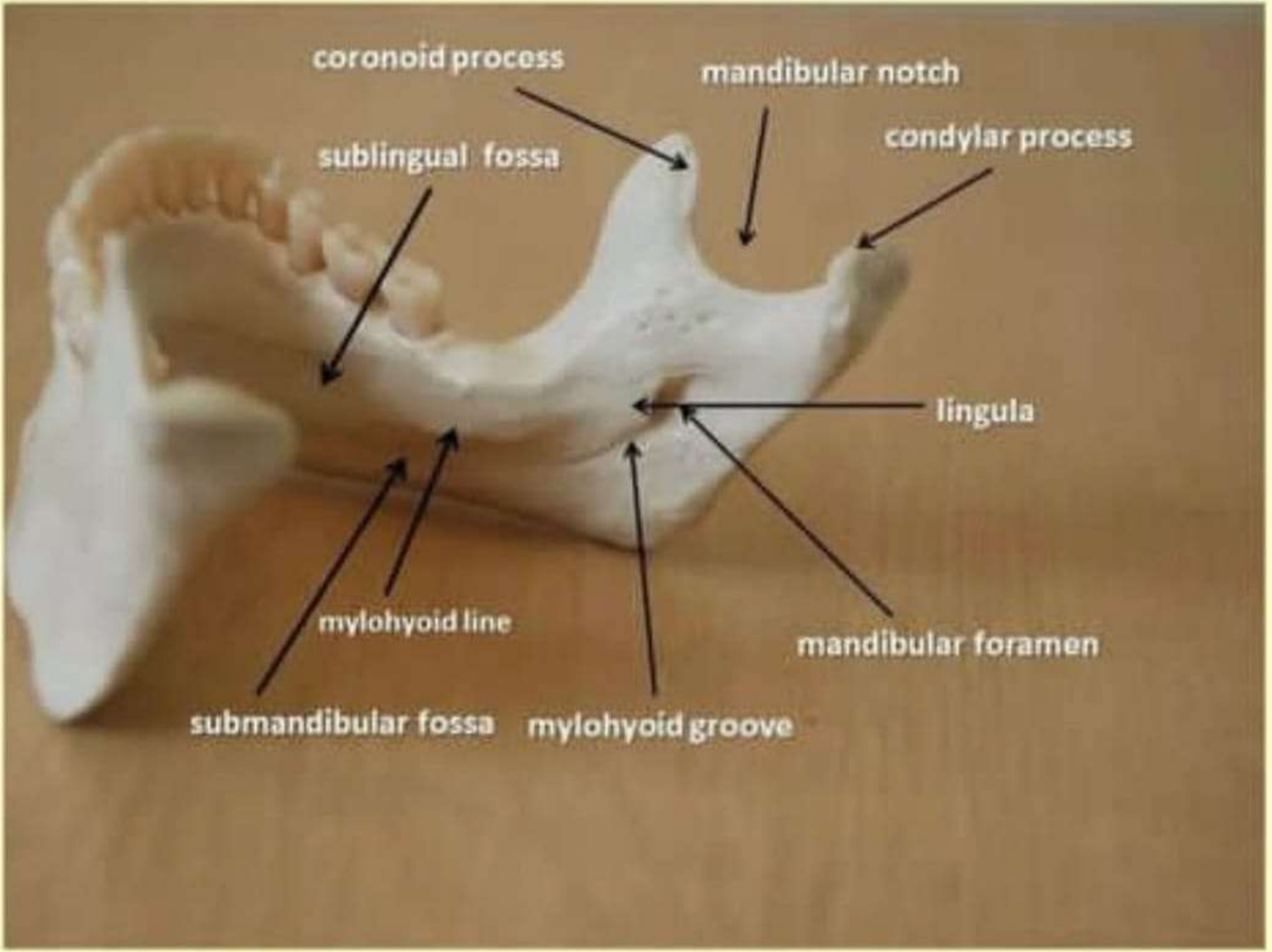
ACCESSORY ROOTS



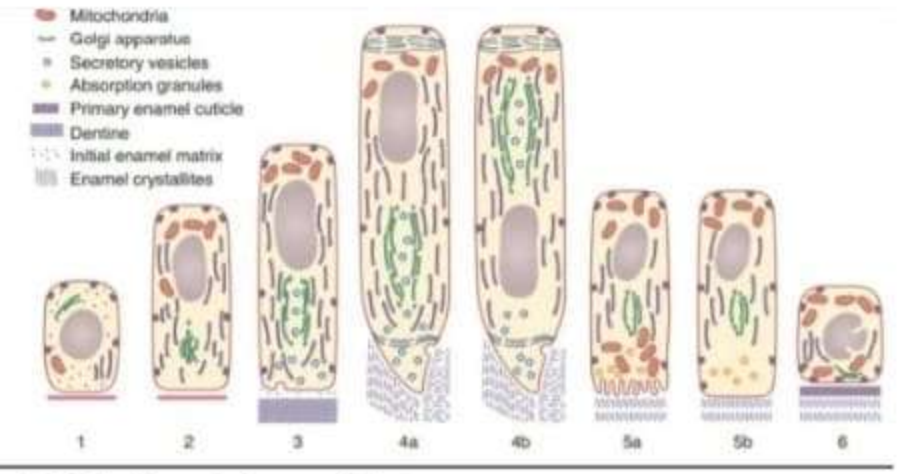
HYPER-CEMENTOSIS

Tooth Eruption
Permanent

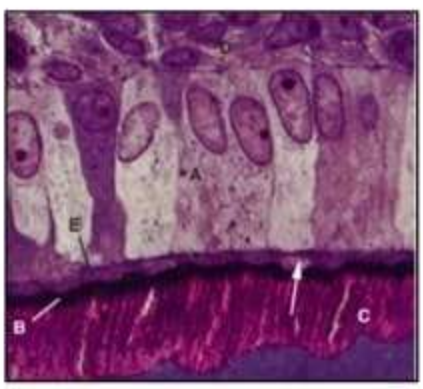



coronoid process
mandibular notch
condylar process
sublingual fossa
lingula
mylohyoid line
mandibular foramen
submandibular fossa
mylohyoid groove

Amelogenesis



g. 22.1 The life cycle of an ameloblast.



A- Ameloblast
B- Enamel Matrix
C- Dentine
D- Stratum Intermedium
E- Terminal bar apparatus

Explanation: The arrowhead shows Tome's process giving the mineralizing front of enamel a picket-fence appearance.

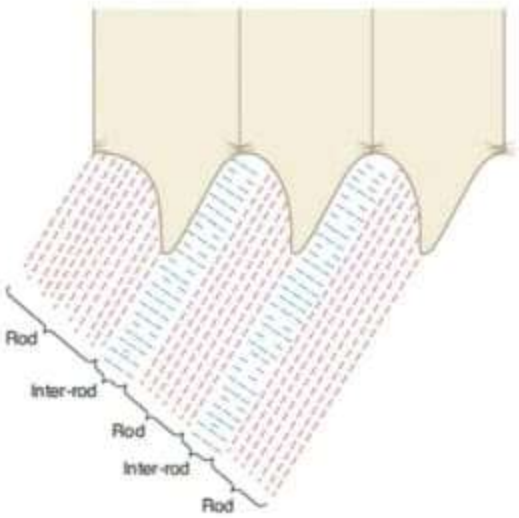


Fig. 22.12 The relationship between Tomes processes and enamel prism formation

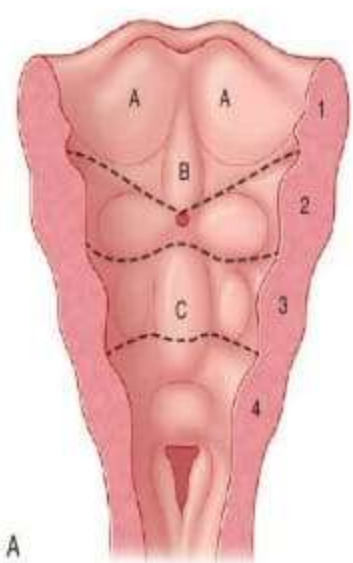






Amelogenesis Imperfecta | Tetracycline Staining | Flourosis

Development of Tongue



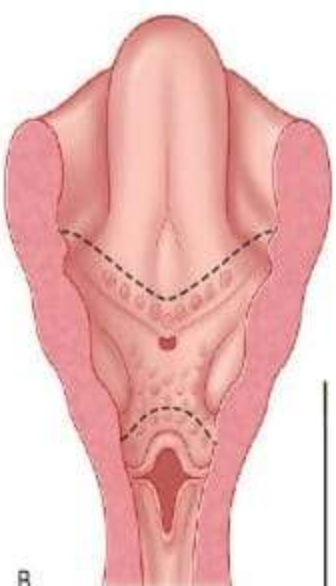
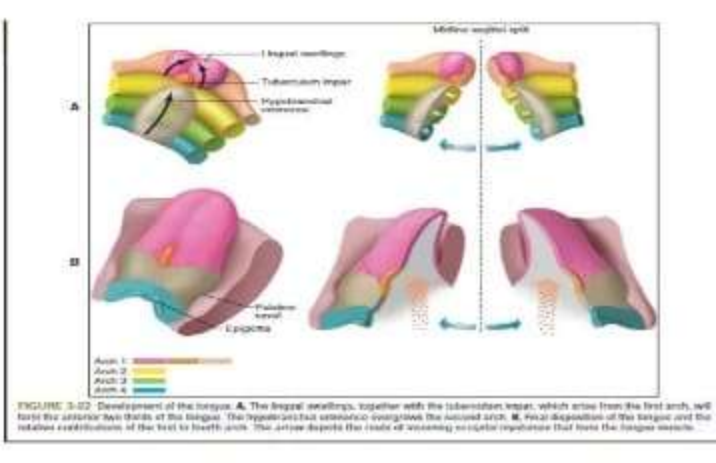


Fig. 20.1 The schema of tongue primordia. (A) The ventral wall of the pharynx in the 4th week of intra-uterine life. (B) The developing tongue at the 5th month. Numbers 1–4 indicate the positions of the pharyngeal (branchial) arches. Four swellings are seen: two distal tongue buds (lateral lingual swellings, A), the median tongue bud (tuberculum impar, B) and the hypopharyngeal eminence (C).



Development of Face

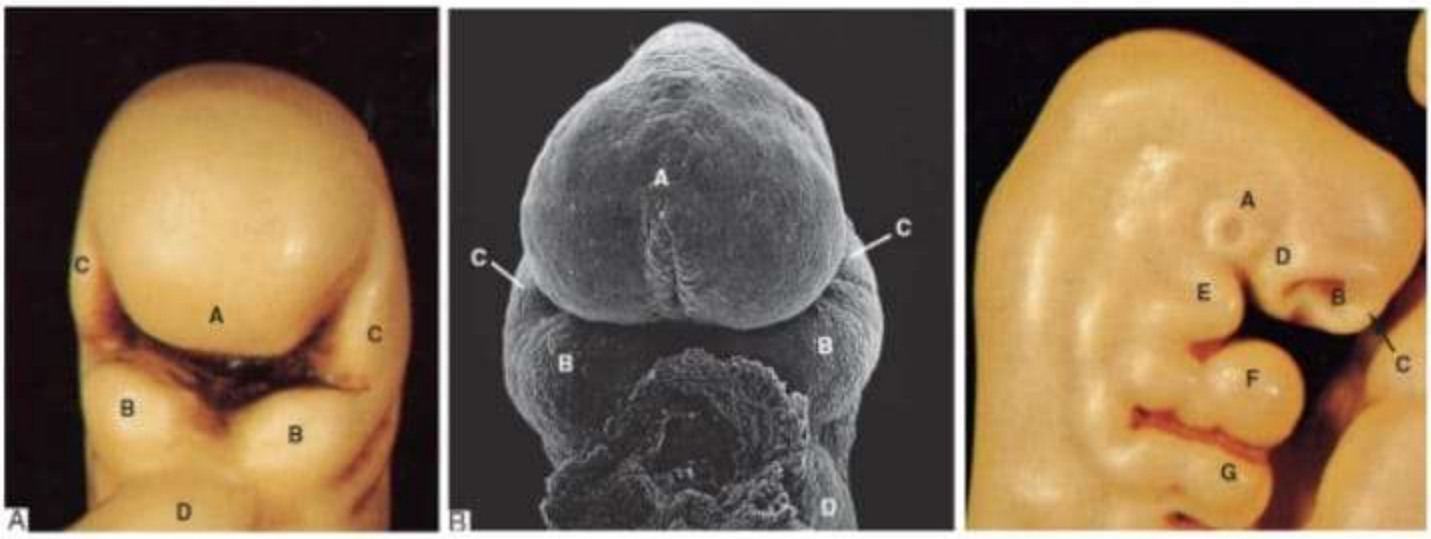
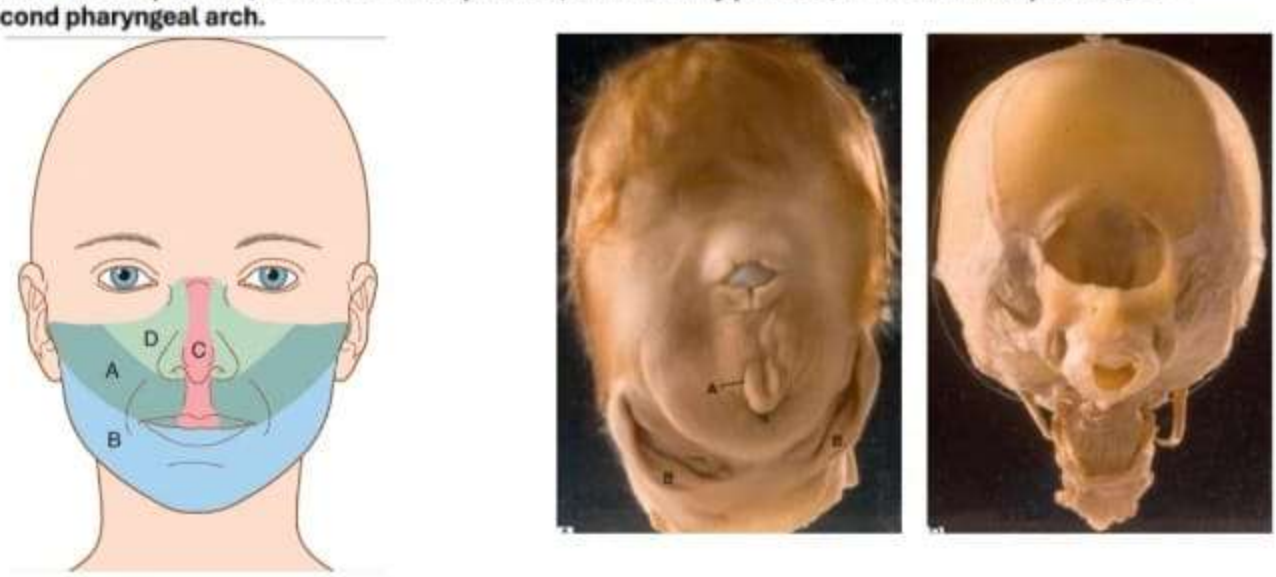


Fig. 17.1 (A) Model of a face of a 4-week-old human embryo (frontal aspect). (B) SEM of the face at an equivalent stage to 4 weeks in the rat (frontal aspect). A = frontonasal process; B = mandibular processes; C = maxillary processes; D = pericardial swelling.

Fig. 17.2 Model of a face of a 5-week-old human embryo (lateral aspect). A = optic placode; B = nasal pit; C = medial nasal process; D = lateral nasal process; E = maxillary process; F = mandibular process; G = second pharyngeal arch.



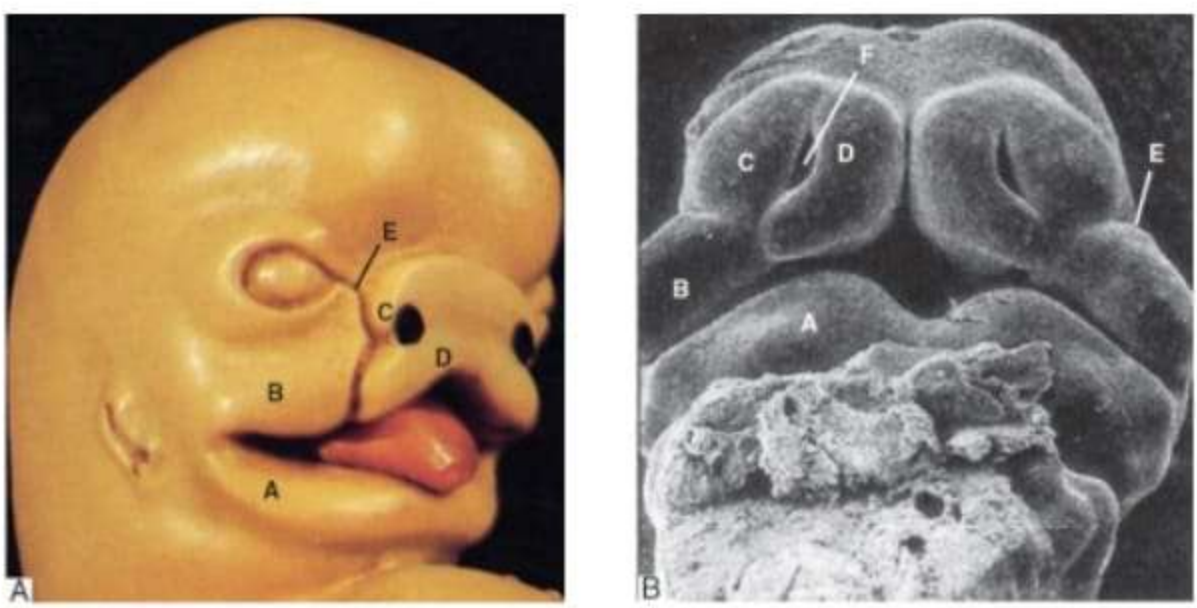
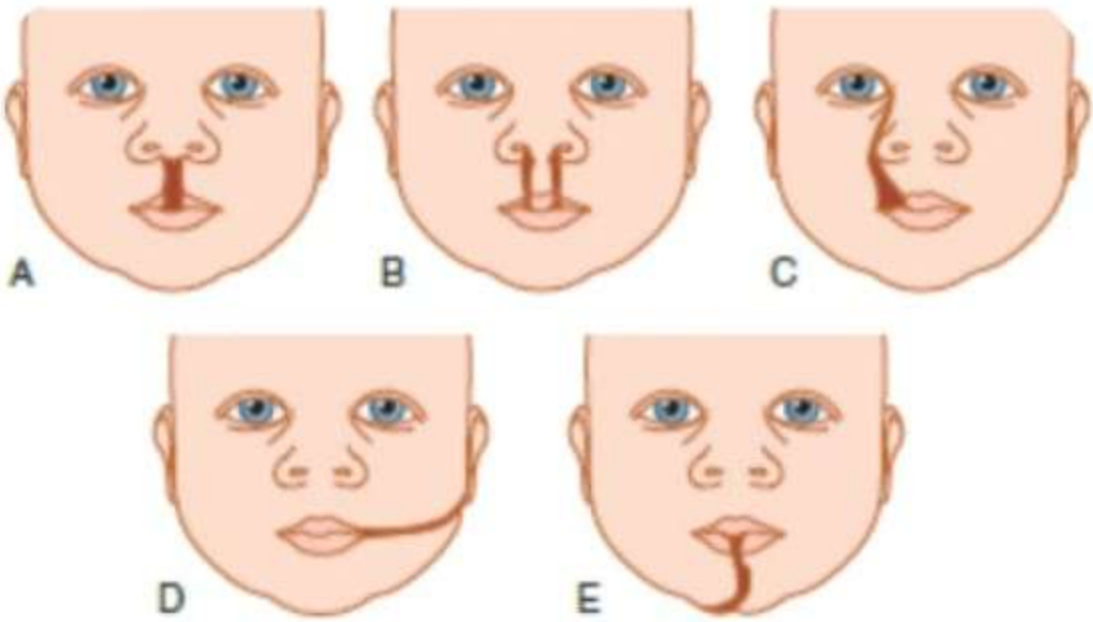
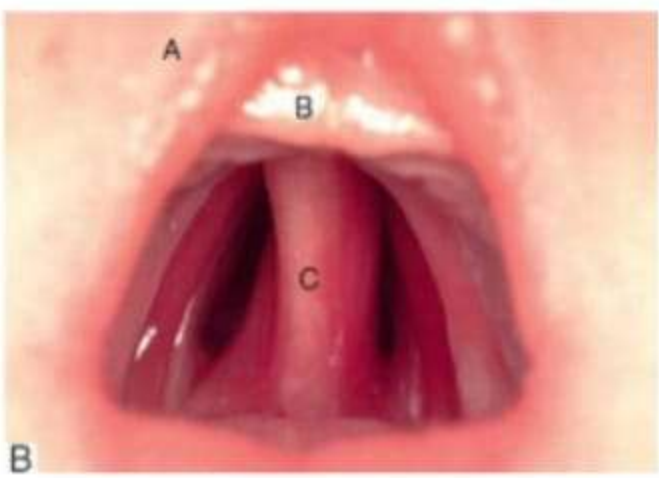


Fig. 17.3 (A) Model of a face of a 6-week-old human embryo (lateral aspect). A = mandibular process; B = maxillary process; C = lateral nasal process; D = medial nasal process; E = naso-optic furrow. (B) SEM of developing rat upper jaw and lip at an equivalent stage to 6 weeks showing the merging maxillary and medial nasal processes.

Development of Face







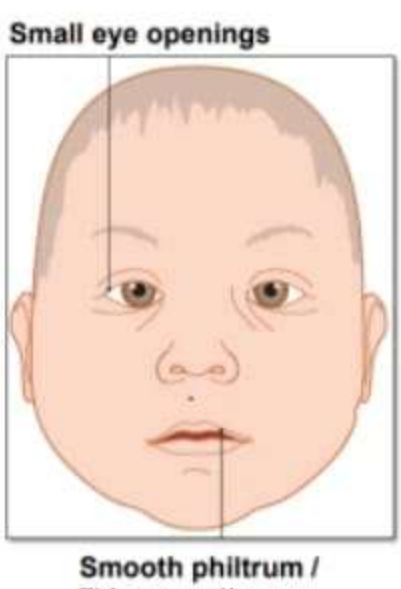


Fig. 17.12 Classic features of fetal alcohol spectrum disorder.



Fig. 17.10 Bilateral cleft lip.



Fig. 17.9 Unilateral cleft lip.

Development of Palate

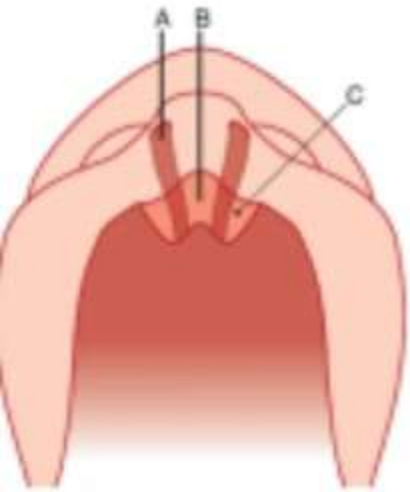


Fig. 18.1 The state of development of the palate by the 6th week of intrauterine life.
A – Primitive nasal cavities: Early right and left nasal sacs.
B – Primary nasal septum: A thin vertical partition formed by the fusion of the medial nasal processes.
C – Primary palate: A small triangular plate of tissue derived from the intermaxillary segment.

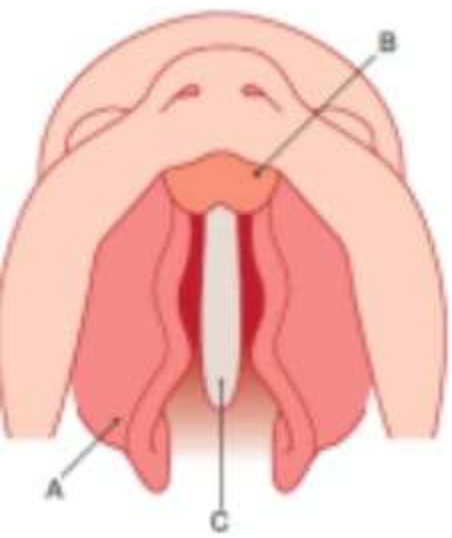
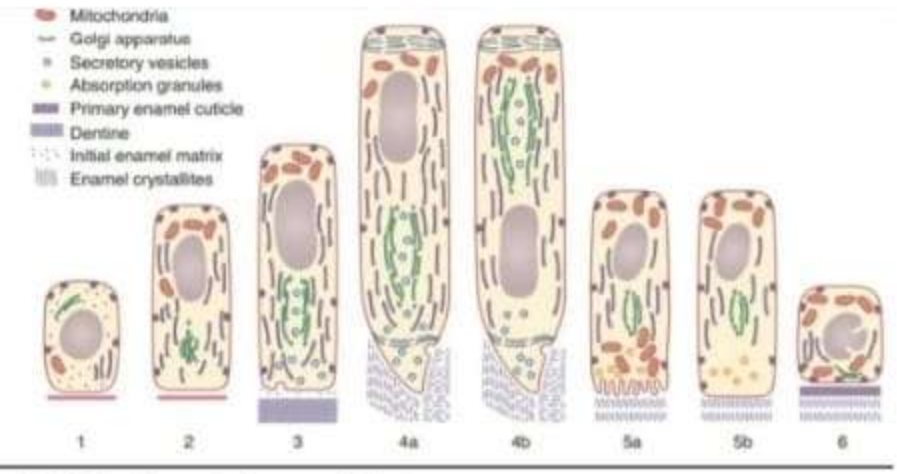


Fig. 18.2 The development of the palate during the 6th week of intrauterine life.
A – Lateral palatal shelves: Outgrowths from the maxillary processes on each side of the tongue. Initially, they grow vertically downward
B – Primary palate: Formed earlier (fusion of medial nasal processes); lies anterior to the incisive foramen.
C – Secondary nasal septum: A downward mesenchymal extension from the frontonasal prominence, which will later fuse with the secondary palate.

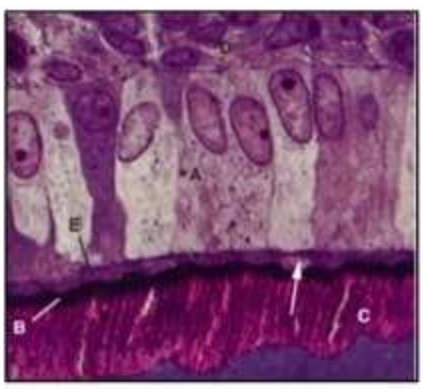


Fig. 18.4 The state of development of the palate during the 8th week of intrauterine life.
A – Palatal shelves: elevate and grow medially and fuse with each other and with the posterior border of the primary palate.
B – Primary palate: lies anterior to the incisive foramen. This fusion creates a Y shaped suture.

Amelogenesis



g. 22.1 The life cycle of an ameloblast.



A- Ameloblast
B- Enamel Matrix
C- Dentine
D- Stratum Intermedium
E- Terminal bar apparatus

Explanation: The arrowhead shows Tome's process giving the mineralizing front of enamel a picket-fence appearance.

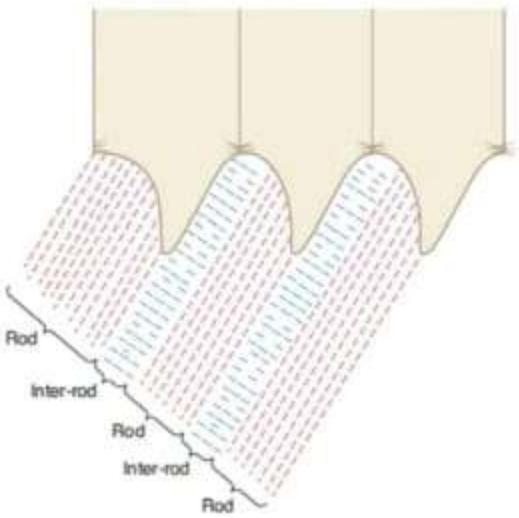


Fig. 22.12 The relationship between Tomes processes and enamel prism formation

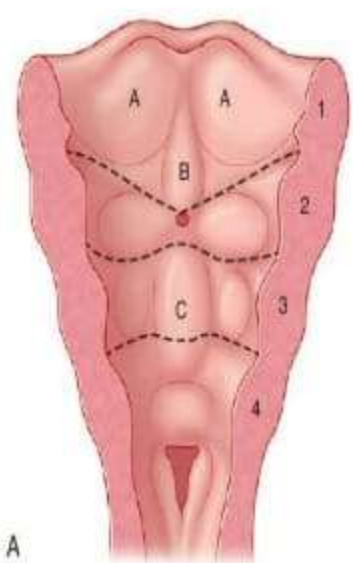






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Development of Tongue



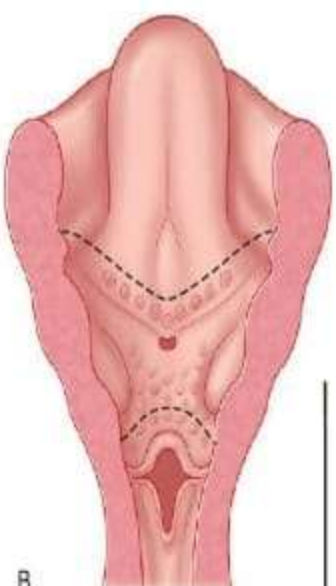
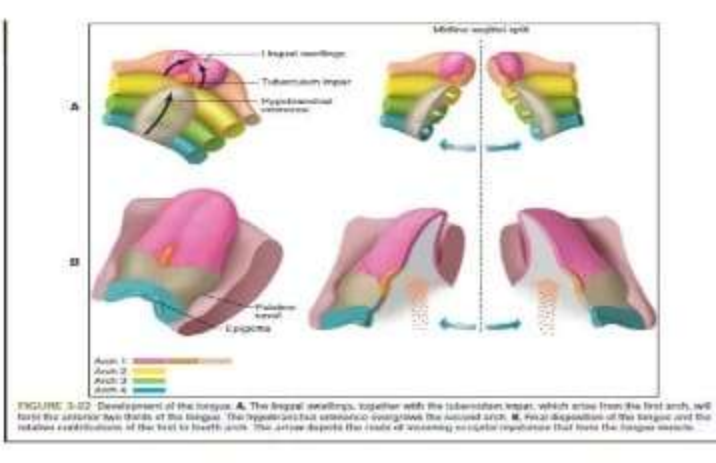


Fig. 20.1 The schema of tongue primordia. (A) The ventral wall of the pharynx in the 4th week of intra-uterine life. (B) The developing tongue at the 5th month. Numbers 1–4 indicate the positions of the pharyngeal (branchial) arches. Four swellings are seen: two distal tongue buds (lateral lingual swellings, A), the median tongue bud (tuberculum impar, B) and the hypopharyngeal eminence (C).

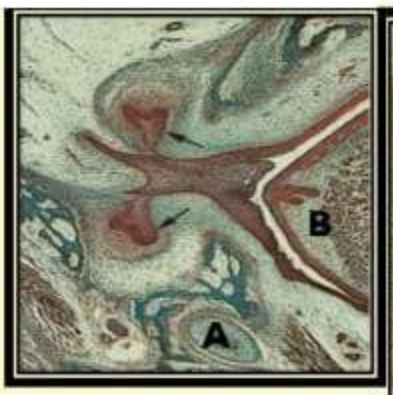


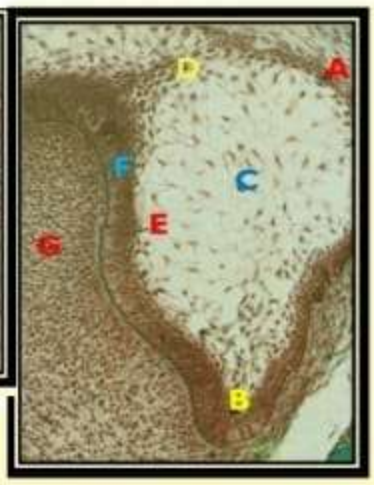
Early Tooth Development

Early Cap Stage of Tooth Development

A : Meckel's Cartilage
B: Developing Tongue

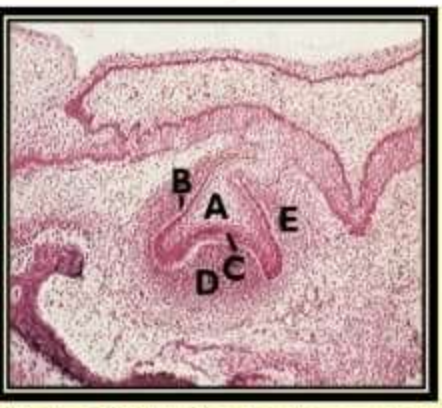
Week : 11th IU (for developing deciduous teeth)





Early Bell Stage of Tooth Development (High-power view)

A : External Enamel Epithelium
B : Cervical Loop
C : Stellate Reticulum
D : Enamel Cord
E : Stratum Intermedium
F : Internal Enamel Epithelium



Late Cap Stage of Tooth Development

A : Stellate Reticulum
B : External Enamel Epithelium
C : Internal Enamel Epithelium
D : Dental Papilla
E : Dental Follicle

Week : 12th IU (for developing deciduous teeth)

Late Bell Stage (Appositional Stage) of Tooth Development

Enamel Matrix
Dentin Matrix

A : permanent tooth

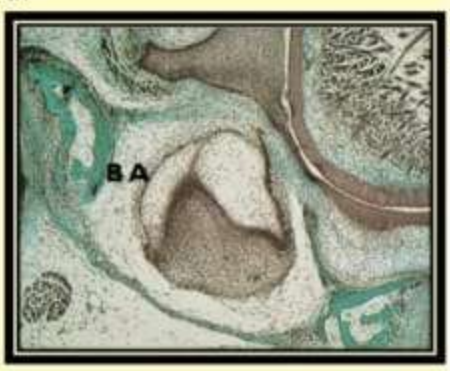


Week : 18th IU (for developing deciduous teeth)

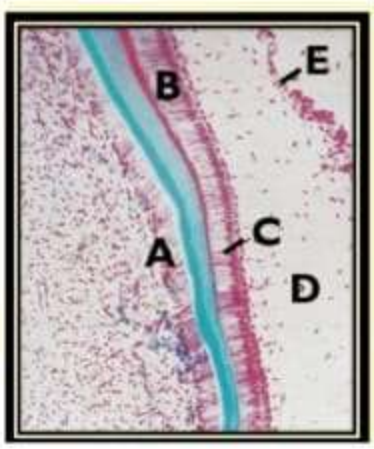
Early Bell Stage of Tooth Development

A : Inner Investing Layer of Dental Follicle

B: Outer Layer of Dental Follicle



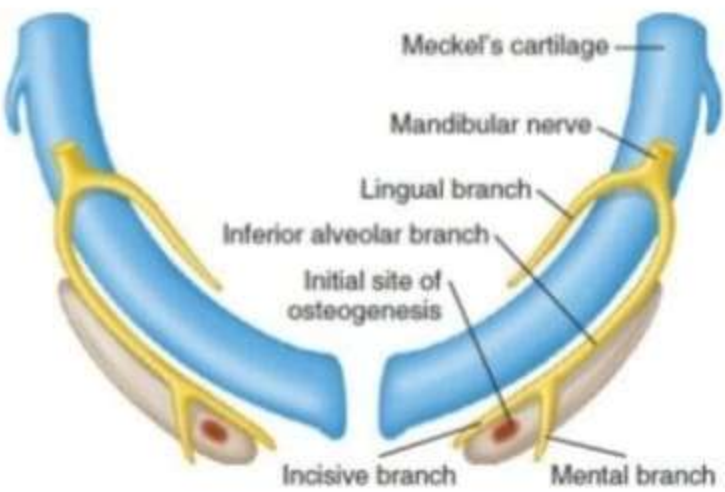
Week : 14th IU (for developing deciduous teeth)



Late Bell Stage (High Power View) of Tooth Development

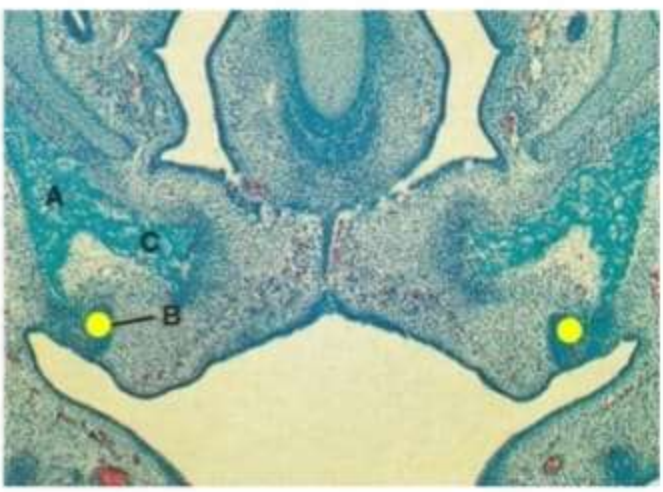
A : Odontoblasts
B : Ameloblasts
C : Stratum Intermedium
D : Stellate Reticulum
E : External Enamel Epithelium

Development of Jaws



DIAGRAMMATIC REPRESNTATION OF EARLY DEVELOPING MANDIBLE:

Shown is site of initial osteogenesis related to mandible formation



CORONAL SECTION THROUGH DEVELOPING ORONASAL REGIONS SHOWING EARLY DEVELOPING MAXILLA:

(A) the region of developing maxilla

(B) The region of developing canine (site of ossification)

(C) palatine process

Dentinogenesis

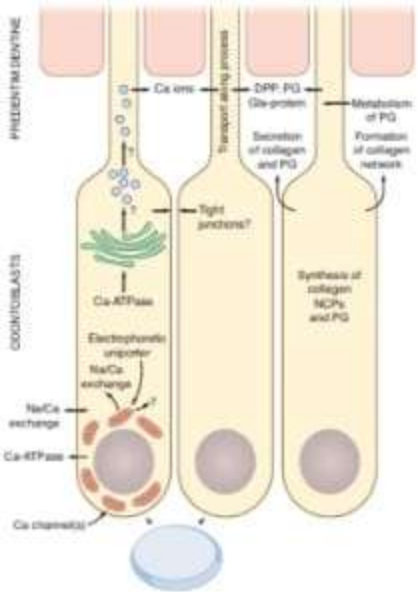


Fig. 23.10. Schematic diagram of fully differentiated active odontoblasts showing different calcium ion transport mechanisms (left side of figure) as well as putative transport routes for the dentine matrix macromolecules (right side of figure).

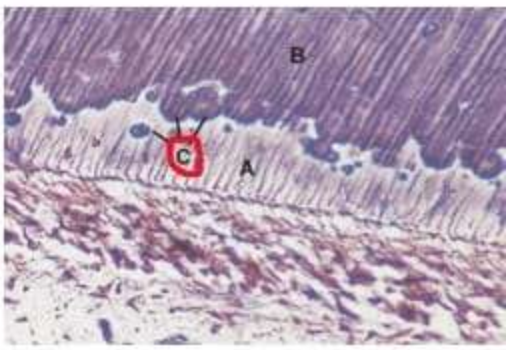


Fig. 23.14 Demineralised section showing the region of mineralising dentine.
A = predentine;
B = mineralized dentine;
C = calcospherites



Fig. 23.7 SEM showing the surface of predentine. The collagen fibrils (brown) form an interlacing network perpendicular to the odontoblast process (blue)



Fig. 23.4 Micrograph of a demineralised section showing 'corkscrew' fibres of von Korff (A). These are first-formed type III and type I collagen fibres whose orientation lie perpendicular from those laid down later.



Fig. 23.25 A patient with dentinogenesis imperfecta. The teeth are malformed and are dark because of the absence of pulp chambers and the consequent loss of translucency. The enamel is normal but flakes off as it is only weakly attached to the dentine.

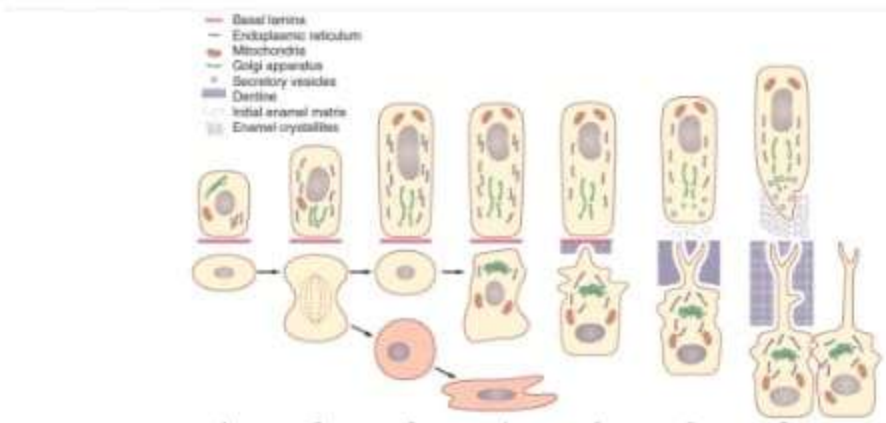


Fig. 23.1 Life cycle of the odontoblast (lower cell line) related to that of the ameloblast (upper cell line). 1 = Ameloblast begins to differentiate first. 2 = Peripheral ectomesenchymal cells divide, with some daughter cells migrating below the odontoblast layer. 3 = Acting on a signal from the ameloblast, the preodontoblasts begin to differentiate. 4 = Synthetic organelles increase in size and number, especially the Golgi apparatus and rough endoplasmic reticulum. 5 = Nucleus moves basally as the cell becomes polarised. A number of odontoblast processes begin to form. One odontoblast process becomes enlarged and begins to secrete matrix. 6 = The odontoblast retreats as matrix is laid down, leaving behind a single main process. Once a narrow layer of matrix is laid down mineralisation commences. 7 = Once the first layer of dentine is laid down the differentiated ameloblast begins to deposit matrix.

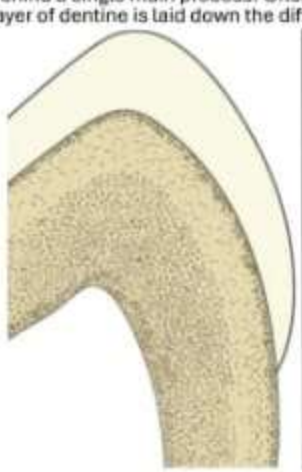
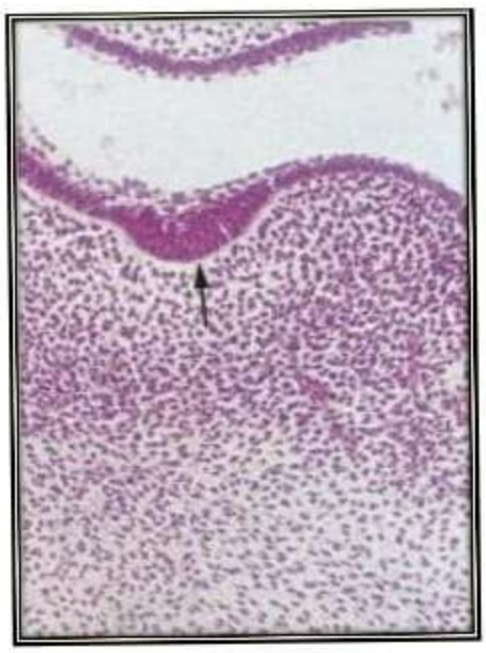
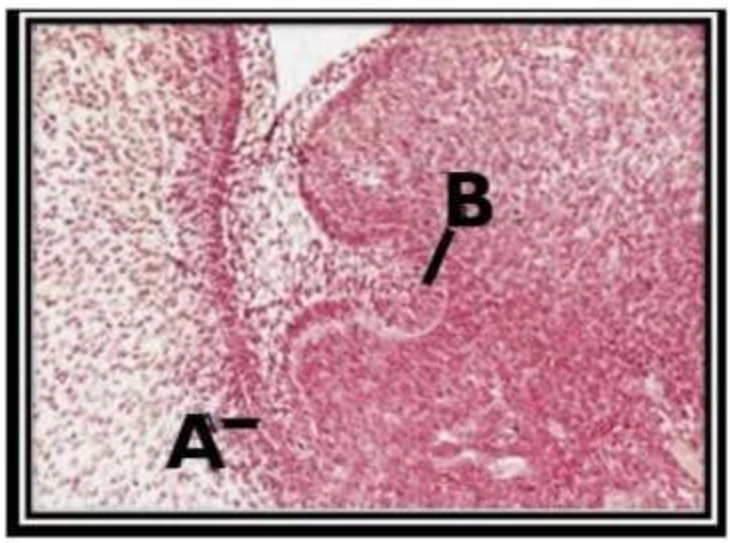


Fig. 23.19 The distribution of calcospherites in dentine. Most calcospherites are spherical. However, many have an arcade shape, such that the round apex of the arcade is directed towards the outer surface of the dentine and the opening is directed towards the pulp. The size varies considerably: the arcade variety tends to be larger than the spherical variety. The size and shape of the calcospherites seem to be governed by the rate of dentine formation and by the rate at which new calcospherites are initiated. There is a fairly consistent pattern of distribution within the tooth. In the mantle dentine of the crown, in the hyaline layer of the root and in the superficial circumpulpal dentine the calcospherites are small, spherical and closely packed. In the middle region of the circumpulpal dentine they are larger, more widely spaced and arcade-like in form (although the region tends to be free of calcospherites towards the root apex). The inner half to two-thirds of the circumpulpal dentine contains spherical calcospherites.

Early Tooth Development





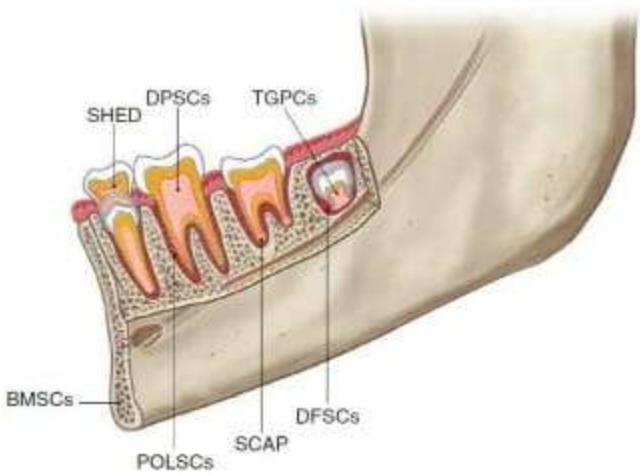
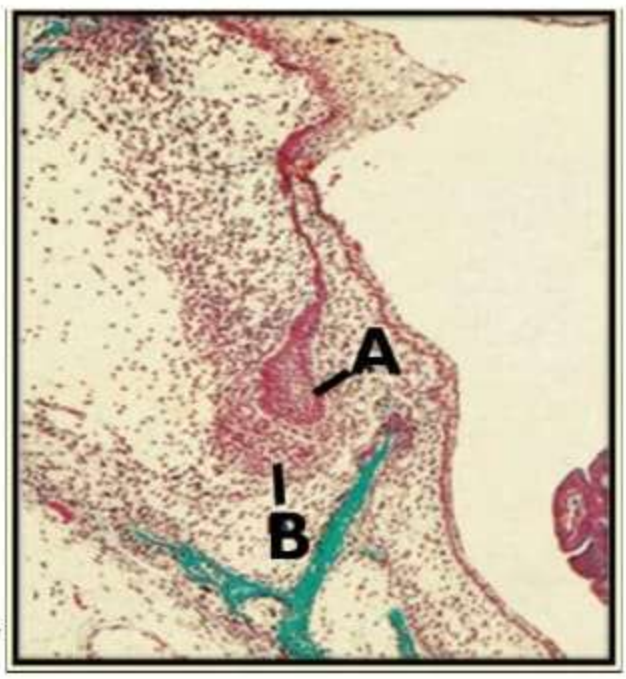


Fig. 21.39 Diagram showing source of stem cells in the regions of the jaws and teeth. Sites of various populations of dental stem cells. BMSC = bone marrow stem cells; DPSC = dental pulp stem cells; SHED = stem cells from the pulp of human exfoliate deciduous teeth; TGPC = stem cells of the tooth germ's papilla cells. Redrawn from Fouad SAA 2015 Dental stem cells: a perspective area in dentistry. *International Journal of Dental Sciences and Research* 3: 15–25.



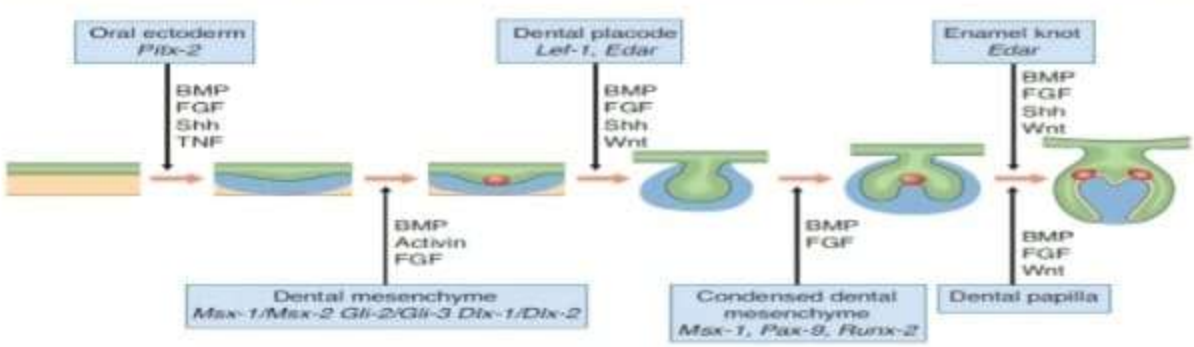


Fig. 21.38 Model of the molecular regulation of tooth development from initiation to crown morphogenesis. Interactions between epithelial (green) and mesenchymal tissues (blue) are mediated by signal molecules (BMP = bone morphogenetic proteins; FGF = fibroblast growth factors; Shh = sonic hedgehog; TNF = tumour necrosis factor). These signals operate throughout development and regulate the expression of genes in the responding tissues (shown in boxes). Signalling centres (red) appear in the epithelium reiteratively and secrete locally many different signals that regulate morphogenesis and tooth shape. Courtesy Prof. I Thesleff.

Genes & Proteins related to different teeth:

Incisor: **MSX1, MSXII, ALX3** (BMP-4)
Molars: **BARX1** (FGF-8, FGF-9)
Canines and premolars: **MSX1, MSXII**

Salivary Glands

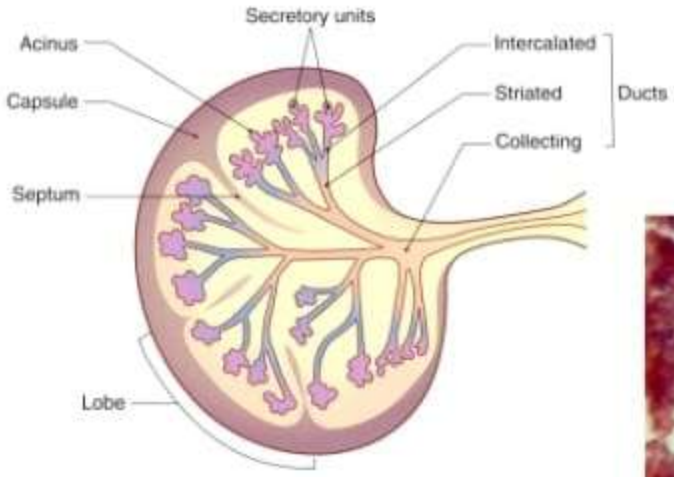


Fig. 16.1 The general organisation of a salivary gland.

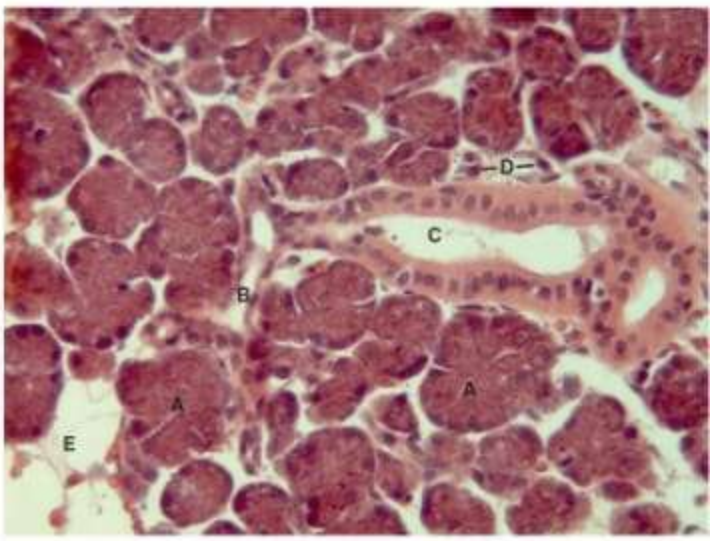


Fig. 16.6 Dark, granular appearance of cytoplasm of serous acini of parotid gland (A). B = intercalated duct; C = striated duct; D = plasma cells; E = fat cell (H & E; ×360). Courtesy Dr J D Harrison.

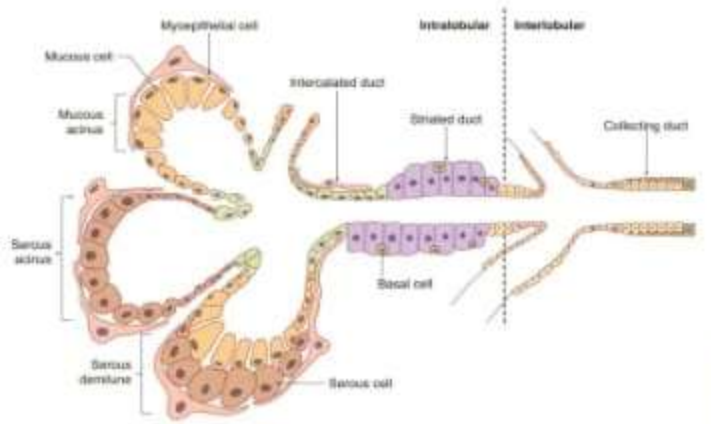


Fig. 16.2 The secretory and ductal elements in a mixed salivary gland. However, as discussed later, the serous demilunes are artefacts of preparation.

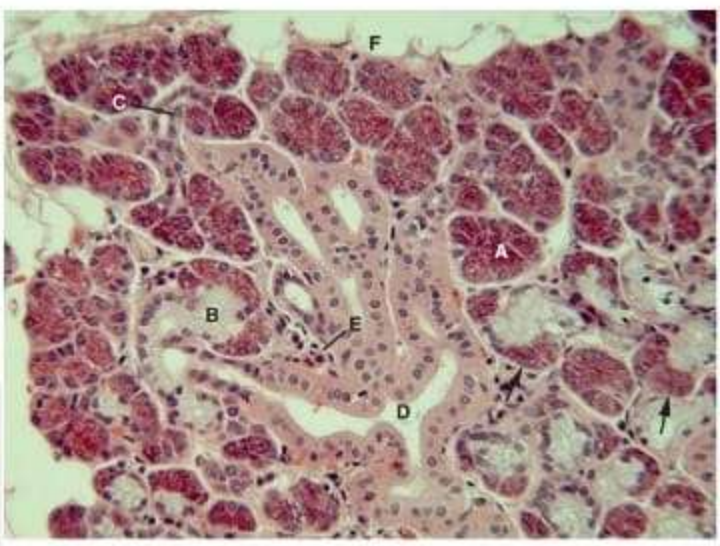
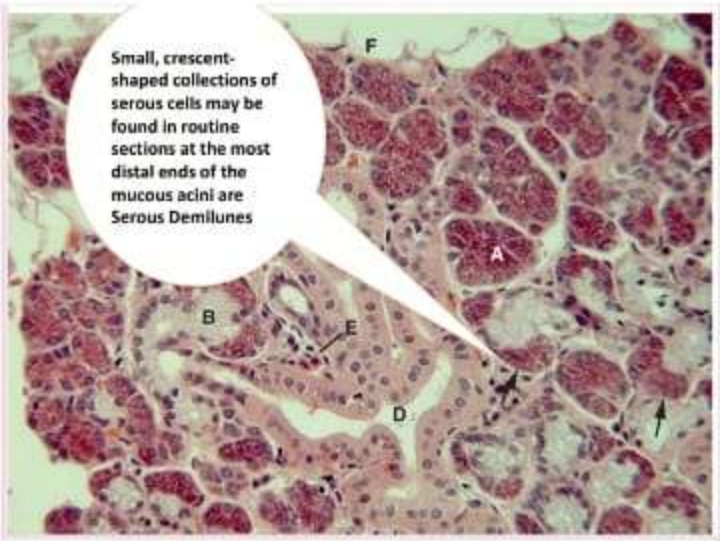


Fig. 16.26 Submandibular gland showing darker serous (A) and lighter mucous (B) acini. Crescent-shaped collections of serous cells form serous demilunes (arrows) (but see text). C = intercalated duct; D = striated duct; E = plasma cells; F = fat cells (H & E; ×245). Courtesy Dr J D Harrison.



Curve of Wilson

- MEDIO-LATERAL CURVATURE OF THE OCCLUSAL PLANE OF POSTERIOR TEETH.
- IT IS ALSO TWO DIMENSIONAL, AND IS AT RIGHT ANGLES TO THAT OF THE CURVE OF SPEE.
- HELPS TO COMPLEMENT THE PATHS OF THE CONDYLES DURING MOVEMENTS OF THE MANDIBLE.
- CROWNS OF MANDIBULAR POSTERIOR TEETH – INCLINE TO THE LINGUAL
- CROWNS OF MAXILLARY POSTERIOR TEETH – INCLINE TOWARD THE BUCCAL.
- BECOMES DEEPER POSTERIORLY, SO THAT THE MOLARS' INCLINATION IS GREATER THAN THAT OF THE PREMOLARS.
- BECAUSE OF THIS CURVE , THE BUCCAL CUSPS OF MANDIBULAR MOLARS, AND THE LINGUAL CUSPS OF MAXILLARY MOLARS USUALLY APPEAR TO BE LONGER.





Enamel → Ectodermal origin

Dentin, cementum, pulp → ectomesenchymal

Enamel is a tissue with no cells within it or it's surface.

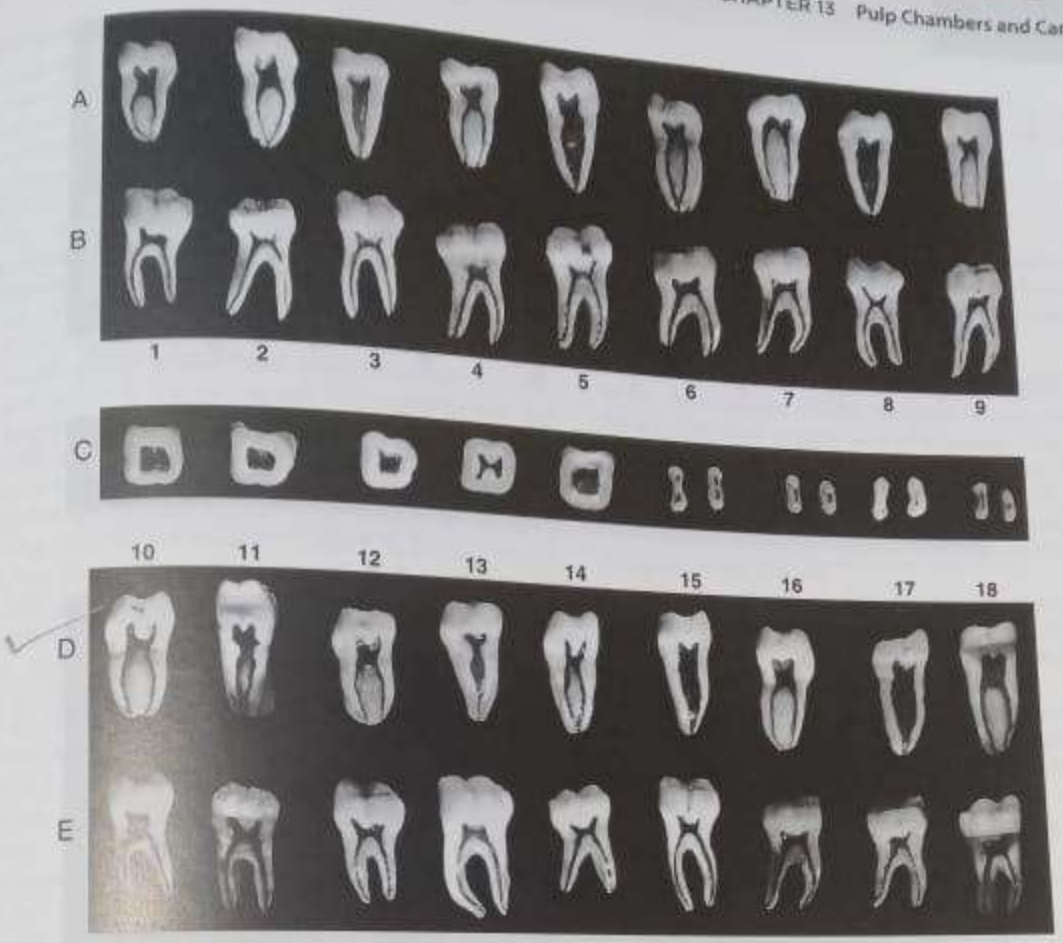
and possibly an apatite, although platelike crystals of octocalcium phosphate have also been reported. The mineral concentration, as measured by X-ray computed microtomography, is significantly higher in sclerotic than in young dentine, consistent with the closure of the tubule lumens. Crystallite size, as measured by small angle X-ray scattering, is slightly smaller in sclerotic dentine. The elastic properties of dentine are unchanged by transparency; however, sclerotic dentine, unlike normal dentine, exhibits almost no yielding before failure. In addition, the fracture toughness is lowered by roughly 20%. In tubules exposed by attrition or caries some occluding components may be derived from saliva.

Dead tracts

If the primary odontoblasts are killed by an external stimulus, or retract before peritubular dentine occludes the tubules, empty tubules will be left. They may be sealed at their pulpal end by tertiary dentine. When ground sections are prepared and mounted, the mounting medium will not enter these sealed-off tubules and they will remain air-filled. Under the microscope, transmitted light will be totally internally reflected and these tubules will appear dark (Figs 9.62, 9.63). This appearance, which is due partly to the pulpal response and partly to the preparatory procedure, has been termed a 'dead tract'.







• Fig. 13.22 Mandibular first molar. (A) Buccolingual section, exposing the mesial or distal aspect of the pulp cavity. This aspect does not appear on the dental radiographs. (B) Mesiodistal section, exposing the buccal or lingual aspect of the pulp cavity. (C) Five cross sections at the cervical line and four cross sections at the midroot. (D) Buccolingual section, exposing the mesial or distal aspect of the pulp cavity. (E) Mesiodistal section, exposing the buccal or lingual aspect of the pulp cavity.

to be located at the tip of the root (see Fig. 13.22B, *1*; E, *10, 11, 12, 14, 15,* and *18*).

Cervical Cross Section (see Fig. 13.22C, 1 to 5)

The cervical cross section of the mandibular first molar is generally quadrilateral in form. Distally, it tapers a little from the wider buccolingual measurement of the mesial aspect of the tooth. The pulp chamber's outline generally follows that of the root (see Fig. 13.22C, *1, 2,* and *5*) but may show buccal (see Fig. 13.22C, *3*) and/or lingual (see Fig. 13.22C, *4*) projections of dentin if the pulp chamber is excessively narrowed by secondary or reactive osteodentin.

The pulp chamber's floor has two small funnel-shaped openings into the mesial root (one buccal and one lingual), whereas the distal aspect of the pulp chamber usually shows a single opening that is less constricted.

Midroot Cross Section (see Fig. 13.22C, 6 to 9)

The midroot view of the mandibular molar usually demonstrates the root canal form, which is consistent with the major form of this tooth.

The mesial root usually is somewhat kidney-shaped, with two separate canals (see Fig. 13.22C, *7* and *9*), but a figure-eight shape of the root is also very common (see Fig. 13.22C, *6* and *8*). The two canals may be totally separate (see Fig. 13.22C and *8*), or one may be confluent with the other canal (see Fig. 13.22C, *7* and *9*). Even three canals may be found in this root on occasion.

The distal root is usually rounder than the mesial root (see Fig. 13.22C, *7* and *9*), but a very wide root is also common (see Fig. 13.22C, *6* and *8*). Those roots that tend to be round usually demonstrate only one canal, whereas the broader oval roots tend to have two canals (see Fig. 13.22C, *6* and *8*) or a very thin canal that is single; they may also possess a dentinal island (see Fig. 13.22C, *8*). Even in a root with a single canal, the distal canal tends to show a developmental depression or concavity on the mesial aspect of the root (see Fig. 13.22C, *7* and *9*).

Mandibular Second Molar

Anatomically, the mandibular second molar has many similarities with the mandibular first molar (Fig. 13.23). The proportions of the crown and root are very similar to those of the mandibular first molar. The roots of the second molar may be straighter with less divergence from the furcation than in the first molar. The roots may be shorter, but there is no assurance that any of these differences will be manifested in any one tooth.

of dentine matrix, fibronectin products and bone morphogenetic protein. It might be envisaged that the addition of suitable activating substances to the region of pulp exposures might aid and speed up clinically the formation of repair dentine over pulp exposures. Recent work however, also suggests that healing can occur naturally providing that the cavity is completely sealed at its margins.

Sclerotic dentine

In addition to infilling with peritubular dentine as a physiological response to ageing (eventually forming translucent dentine – see pages 155–157), dentinal tubules commonly fill in as a response to an external stimulus such as under slowly advancing caries or beneath areas of severe attrition (Figs 9.59, 9.60). This type of dentine is termed sclerotic dentine and, like translucent dentine, will present as areas that lack structure and appear transparent (Fig. 9.61). Little is known about the precipitated material but it appears to differ from peritubular dentine and is thus thought not to be formed by the odontoblast. The mineral is crystalline

Station No.35



05 Type of tooth section is:

a) BL
b) ML
c) DL
d) BD
e) Distal

06 Number of pulp canals present in this tooth are:

a) One
b) Two
c) Three
d) Four
e) Five



75 The oblique facial cleft results from lack of fusion between?

a. Two medial nasal processes
b. Maxillary process and median nasal process
c. Medial nasal and lateral nasal process
d. Maxillary process and lateral nasal process

d

76 Rubella virus contributes in facial and palatine clefts in classic example of?

a. Xray radiation
b. Hormones
c. Drugs
d. Infectious agents

d

Station no. 28



55 Mandible is formed from
a. Reichets's Cartilage
b. Meckel's Cartilage
c. Mandibular processes
d. All of the above

b

56 The marked structure A and B are ?
a. Meckel's cartilage and mandibular nerve
b. Mandibular nerve and lingual nerve
c. Inferior alveolar nerve and incisive branch
d. Mandibular nerve and mental nerve

a



65 Intraoral view of dentition shows which dental malformation?

a. AmelogenesisImperfecta
b. DentinogenesisImperfecta
c. Osteogenesis Imperfecta
d. Discoloration of Mandibular teeth

a

66 Mentioned malformation is ?

a. Autosomal recessive defect
b. Autosomal dominant defect
c. It is due to stains in foods
d. It is caused by certain infections

c

Station No. 37



73. Identify the labeled structure B ?

A. Circumvillate papillae
B. Fordyce's spots
C. Parotid gland
D. Taste bud

17 / 32

74. main function of striated excretory duct is?

A. Modification of primary saliva
B. Modification of secondary saliva
C. Regulation of flow of saliva
D. Inclusion of minerals and modification of saliva

Station No.36



Q1 Identify the tooth:

a) Mandibular 2nd premolar mesial aspect
b) Maxillary 1st premolar mesial aspect
c) Mandibular 1st premolar mesial aspect
d) Maxillary 1st premolar distal aspect
e) Maxillary 2nd premolar mesial aspect

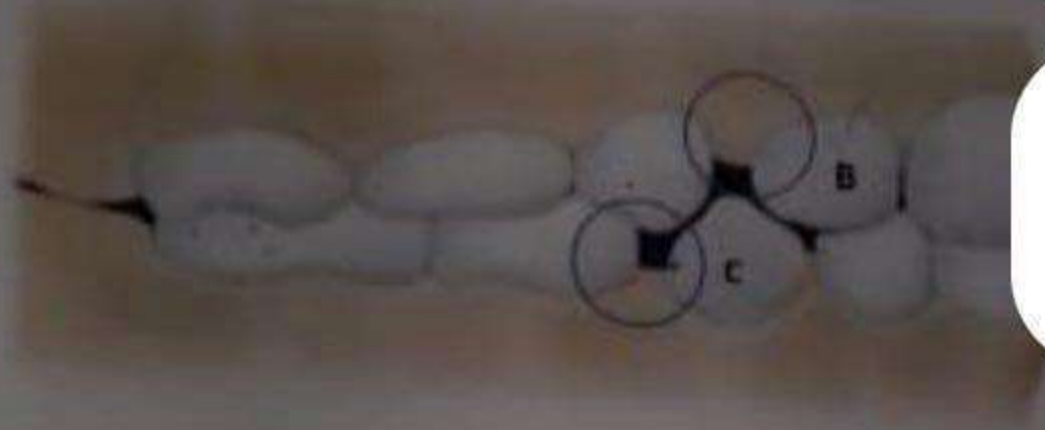
b

Q2 Two roots are found in:

a) Maxillary [illegible] premolar
b) Mandibular [illegible] premolar
c) Maxillary 1st premolar
d) Maxillary 2nd premolar
e) Primary canine

C

Station no. 34



18 / 32

67 Primate space is

a) Naturally occurring spacing between primary dentition.
b) Artificially occurring spacing between primary dentition.
c) Naturally occurring spacing between permanent dentition
d) Artificially occurring spacing between permanent dentition

a

68. In maxillary arch primate space is located in between:

a) Central Incisor and Lateral Incisor
b) Lateral Incisor and canine
c) Canine and Premolar
d) Premolar and molar

b



83. Identify the spot (A)
a) Owen contour of lines
b) Tome's granular layer
c) Interglobular dentine
d) Lines of von Ebner.

C

84 Mineralization pattern of dentine is:
a) Hypo mineralized lines
b) Globular Dentine
c) Lines of Von Ebner
d) Owen Contour lines

?

Station no. 27



19 Identify the marked structure

a) Lingual ridge
b) Mesial marginal ridge
c) Distal marginal ridge
d) Lingual surface
e) Lingual elevation

21 / 32

20 According to sequence of eruption, this tooth is the

a) Last succedeous tooth in the oral cavity
b) First succedeous tooth in the oral cavity
c) second succedeous tooth in the oral cavity
d) third succedeous tooth in the oral cavity
e) Second Last succedeous tooth in the oral cavity

Station No. 40



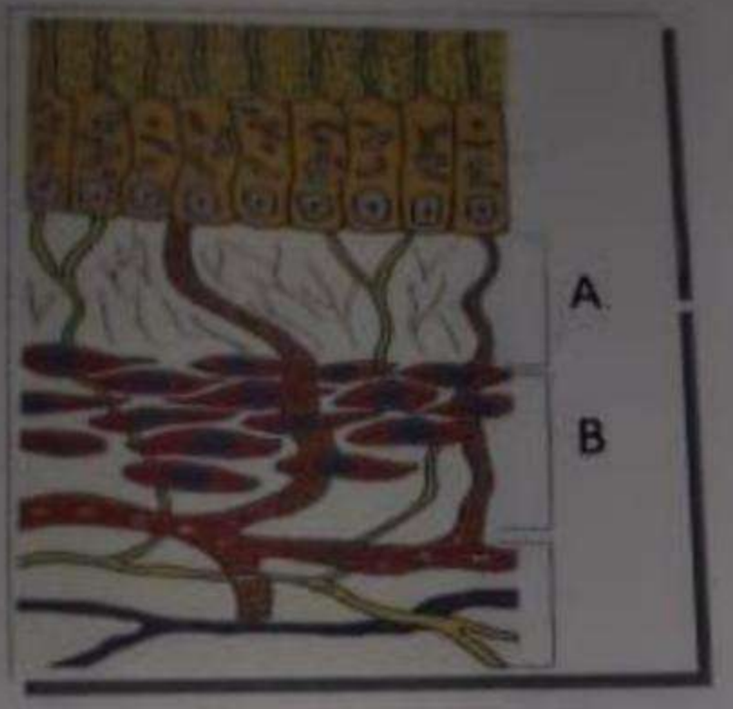
79 Identify the spot A
a) Intra tubular dentine
b) Intertubular dentine
c) Peritubular dentin
d) Secondary dentine

?

80 Identify the spot B.
a) Intra tubular dentine
b) Intertubular dentine
c) Intertubular dentin
d) Secondary dentine

intertubular dentine

Station No.15



17. Identify the spot A
a) Dentine
b) Cell free zone
c) Cell rich zone
d) Pulp core

b

18. Identify the spot B
a) Cell rich zone
b) Pulp core
c) Dentine
d) Cell free zone

a

Station No 18



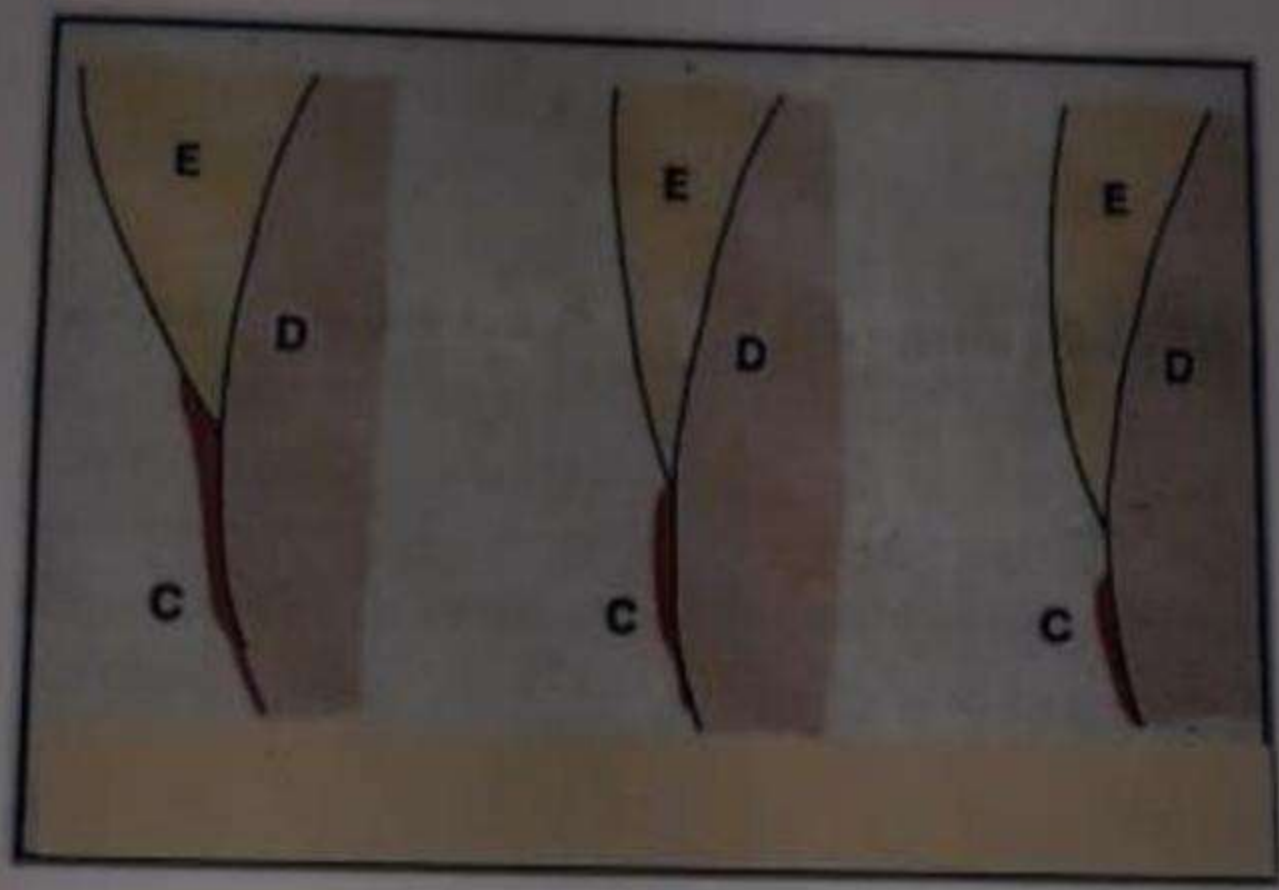
35. Identify the occluasal surface of
    a. Mandibular 1st molar
    b. Maxillary 2nd molar
    c. Maxillary [illegible] molar
    d. Mandibular [illegible] molar

c

36. Identify marked spot
    a. Fifth cusp
    b. Cusp of Carabelli
    c. Distal cusp
    d. Distolingual cusp

b

Station No 22



43 Identify the spot.

a) Types of dentine
b) Types of cementum
c) Types of enamel
d) Types of cemento-enamel junction

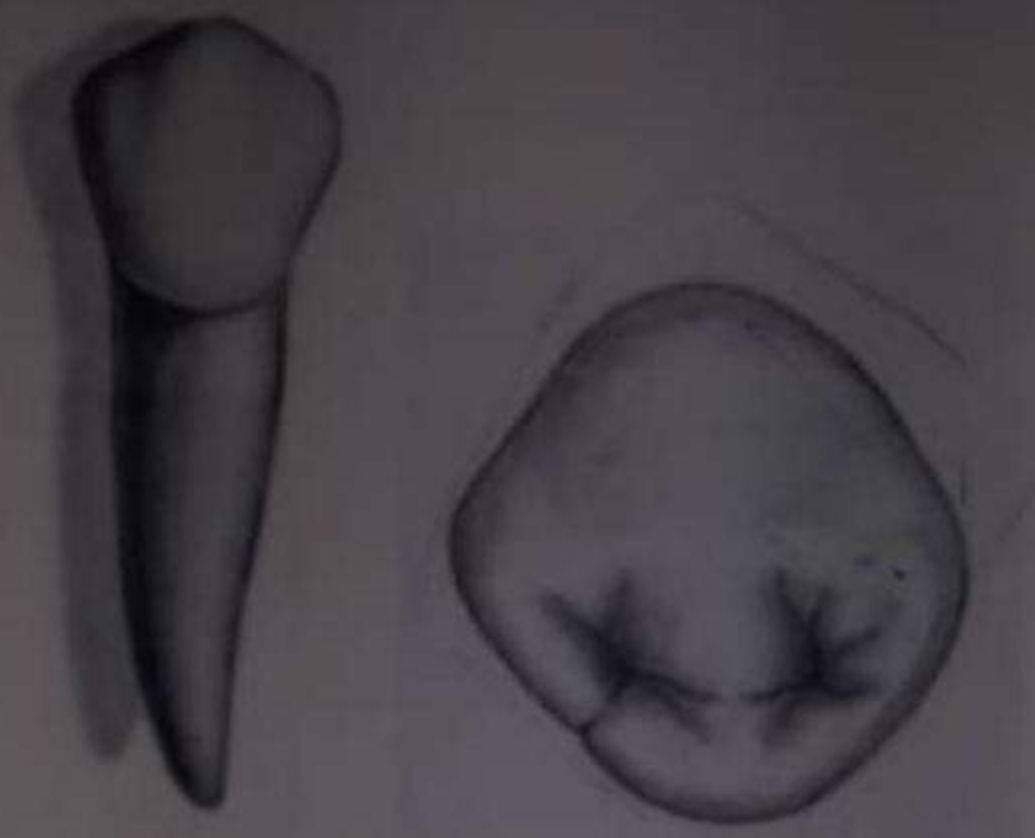
d

44 How many % enamel and dentin fail to meet, leaving dentin exposed ?

a) 5-10%
b) 25-30%
c) 10-20%
d) 15-40%

a

Station No 19



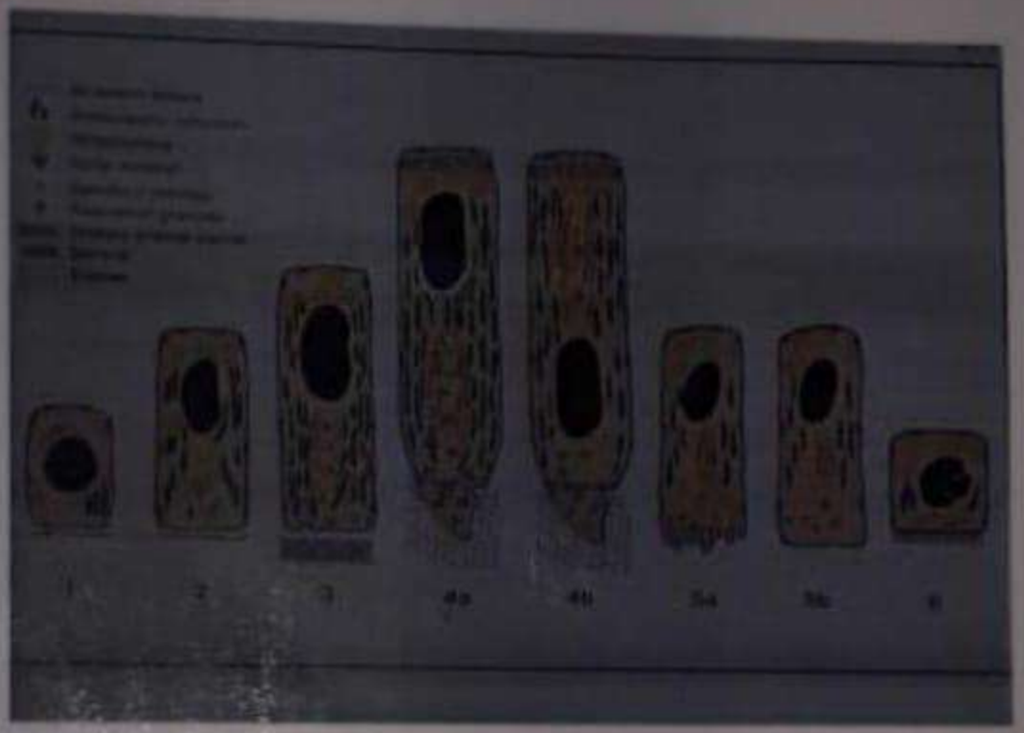
37 Identify the tooth:

a) Mandibular canine
b) Maxillary 2nd premolar
c) Mandibular 2nd premolar
d) Maxillary 1st premolar
e) Mandibular 1st premolar

38 The occlusal surface of mandibular first premolar is roughly:

a) Circular
b) Triangular
c) Diamond sha
d) Spoon shaped
e) Elevated

Station No 30



59 Identify the spot
a) Amelinogenesis
b) Amelogenesis
c) Formative stage of ameloblasts
d) Maturative stage of ameloblasts.

b

60 Amelogenesis occur during :
a) Formative
b) Maturative
c) Morphogenic stage
d) Both A& B

?

Station no 25



49 The malformation in picture is called?

a. Down's Syndrome
b. Treacher Collin Syndrome
c. OsteogenesisImperfect
d. Ehlers – Danlos Syndr

b

50 This malformation is caused due to?

a. Failure in proper development of head and trunk
b. Failure in migration of neural crest cells to facial region
c. Failure in migration of ectomesenchyme cells to facial region
d. Failure in development and migration of neural crest cells

b

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predentin



The innermost layer of dentin is known as **predentin**, and is the initial dentin matrix that is laid down prior to mineralization.

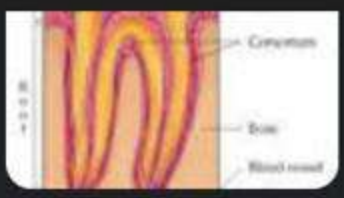
https://en.m.wikipedia.org › wiki

Dentin - Wikipedia

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stage of tooth development. The formation of dentin must always occur before the formation of enamel.



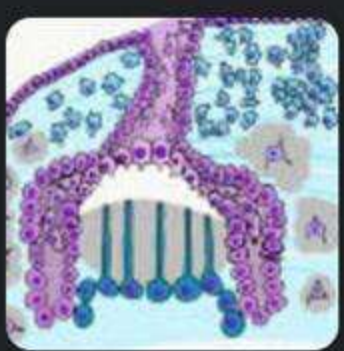
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What is the first thin layer of dentin that is formed?

Mantle dentin forms first, and is a thin layer (15-30mm) closest to the enamel. Mantle dentin contains few dentinal tubules, they are filled in during the maturation stage. Mantle dentin is where globular mineralization occurs. The collagen fibers that remain in mantle dentin run perpendicular to the DEJ. 31-Jul-2021



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More results

Where is primary dentin formed?

in dentine (Fig. 9.31). The most peripheral region beneath the enamel and cementum has special characteristics bestowed on it by being the earliest formed part of the tissue. In the crown this first-formed layer is known as mantle dentine. In the root there are two morphologically recognisable outer zones: the hyaline layer and the granular layer (of Tomes). There is some controversy (discussed below) as to whether the hyaline layer is properly a component of dentine or cementum or is a mingling of components of both tis

Coincident with this deposition of collagen, the plasma membrane of odontoblasts adjacent to the differentiating ameloblasts extends stubby processes into the forming extracellular matrix (see Figure 8-16). On occasion one of these processes may penetrate the basal lamina and interpose itself between the cells of the inner enamel epithelium to form what later becomes an enamel spindle (see Chapter 7). As the odontoblast forms these processes, it also buds off a number of small, membrane-bound vesicles known as *matrix vesicles*, which come to lie superficially near the basal lamina (Figure 8-17; see also Figures 8-12 and 8-16, *A*). The odontoblast then develops a cell process, the odontoblast process or Tomes' fiber, which is left behind in the forming dentin matrix as the odontoblast moves away toward the pulp (see Figure 8-15). The mineral phase first appears within the matrix vesicles as single crystals believed to be seeded by phospholipids present in the vesicle membrane (see Figure 8-16). These crystals grow rapidly and rupture from the confines of the vesicle to spread as a cluster of crystallites that fuse with adjacent clusters to form a continuous layer of mineralized matrix. The deposition of mineral lags behind the formation of the organic matrix so that a layer of organic matrix, called *predentin*, always is found between the odontoblasts and the mineralization front. After mineral seeding, noncollagenous matrix proteins produced by odontoblasts come into play to regulate mineral deposition. In this way coronal mantle dentin is formed in a layer approximately 15 to 20 µm thick onto which then is added the primary (circumpulpal) dentin.

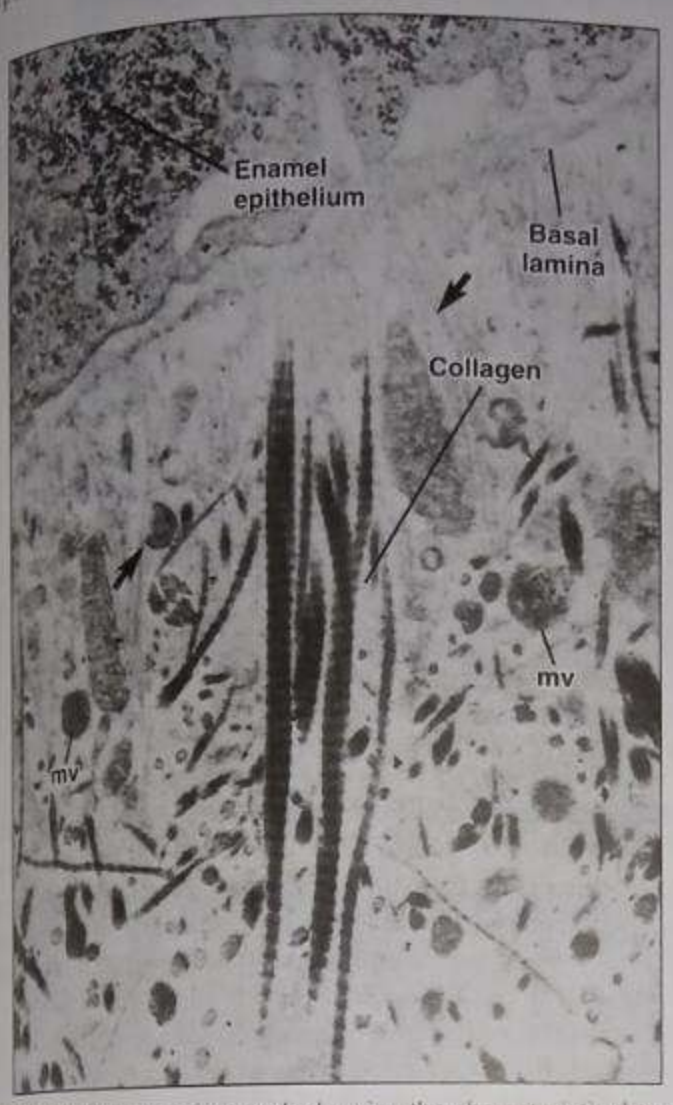


FIGURE 8-12 Electron micrograph showing the characteristic deposition of first collagen fibers to form coronal mantle predentin. Large-diameter collagen fibers *(Collagen)* intermingle with aperiodic fibrils *(arrows)* associated with the basal lamina supporting the enamel epithelium. *mv*, Matrix vesicle. (From Ten Cate AR: A fine structural study of coronal and root dentinogenesis in the mouse: observations on the so-called von Korff fibres and their contribution to mantle dentine. *J Anat* 125:183-197, 1978.)

Vascular Supply

Chapter 1 stated the requirements for good blood supply during the formative phase of hard tissue formation. During dentinogenesis,

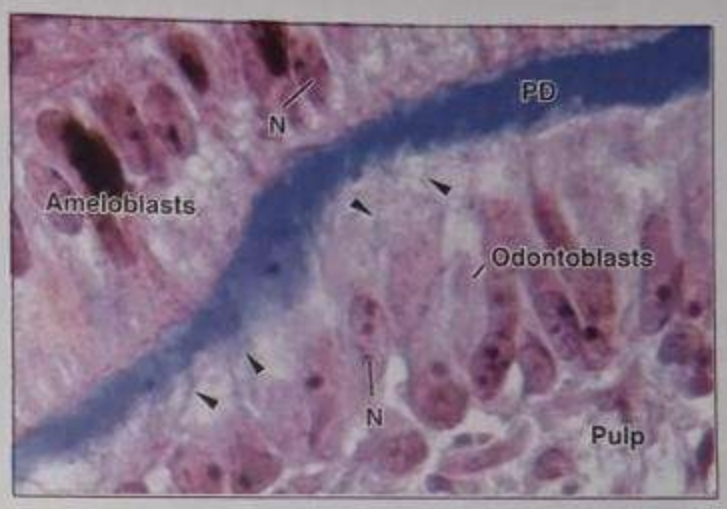


FIGURE 8-14 Light micrograph of a paraffin section specially stained for collagen. Von Korff's fibers appear as convoluted, bluish threadlike structures *(arrowheads)* that originate deep between odontoblasts. *N*, Nucleus; *PD*, predentin.

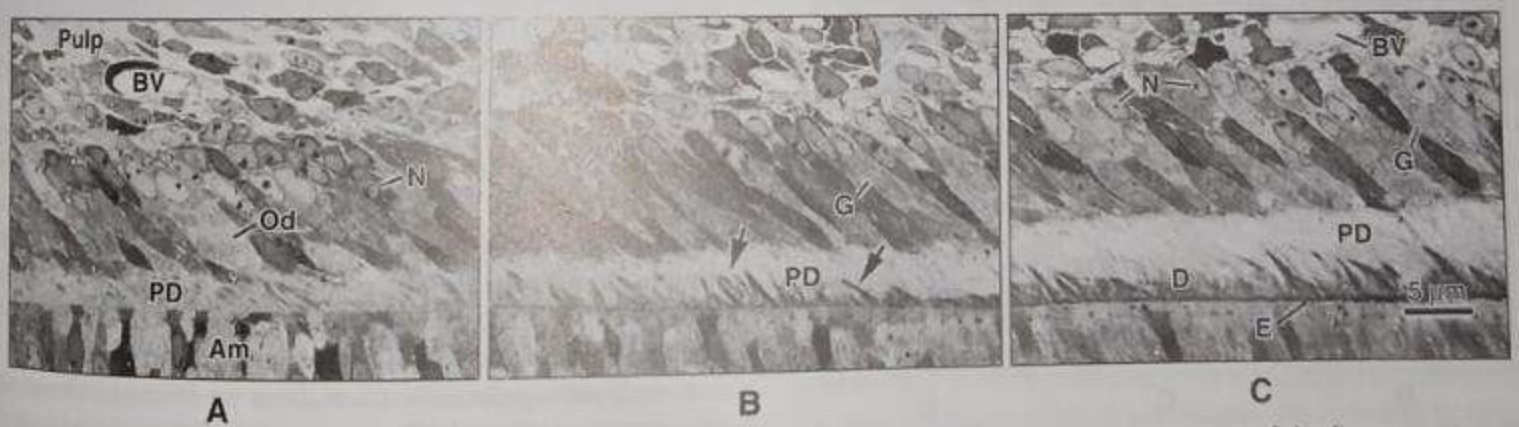


FIGURE 8-13 **A** to **C,** Scanning electron micrographs of tissue sections illustrating the formation of the first layer of (mantle) dentin *(D)* in a rat incisor. Differentiated odontoblasts are tall columnar cells tightly grouped in a palisade arrangement. Their nucleus *(N)* is situated basally, the Golgi complex *(G)* occupies much of the supranuclear compartment, and their body is inclined with respect to that of the ameloblasts *(Am)*. **B,** A concentration of large-diameter collagen fibrils *(arrows)* can be seen in the forming predentin *(PD)* matrix near the surface of the ameloblasts. **C,** As this matrix mineralizes, the fibrils become incorporated in the mantle dentin. *BV*, Blood vessel; *E*, enamel; *Od*, odontoblasts.

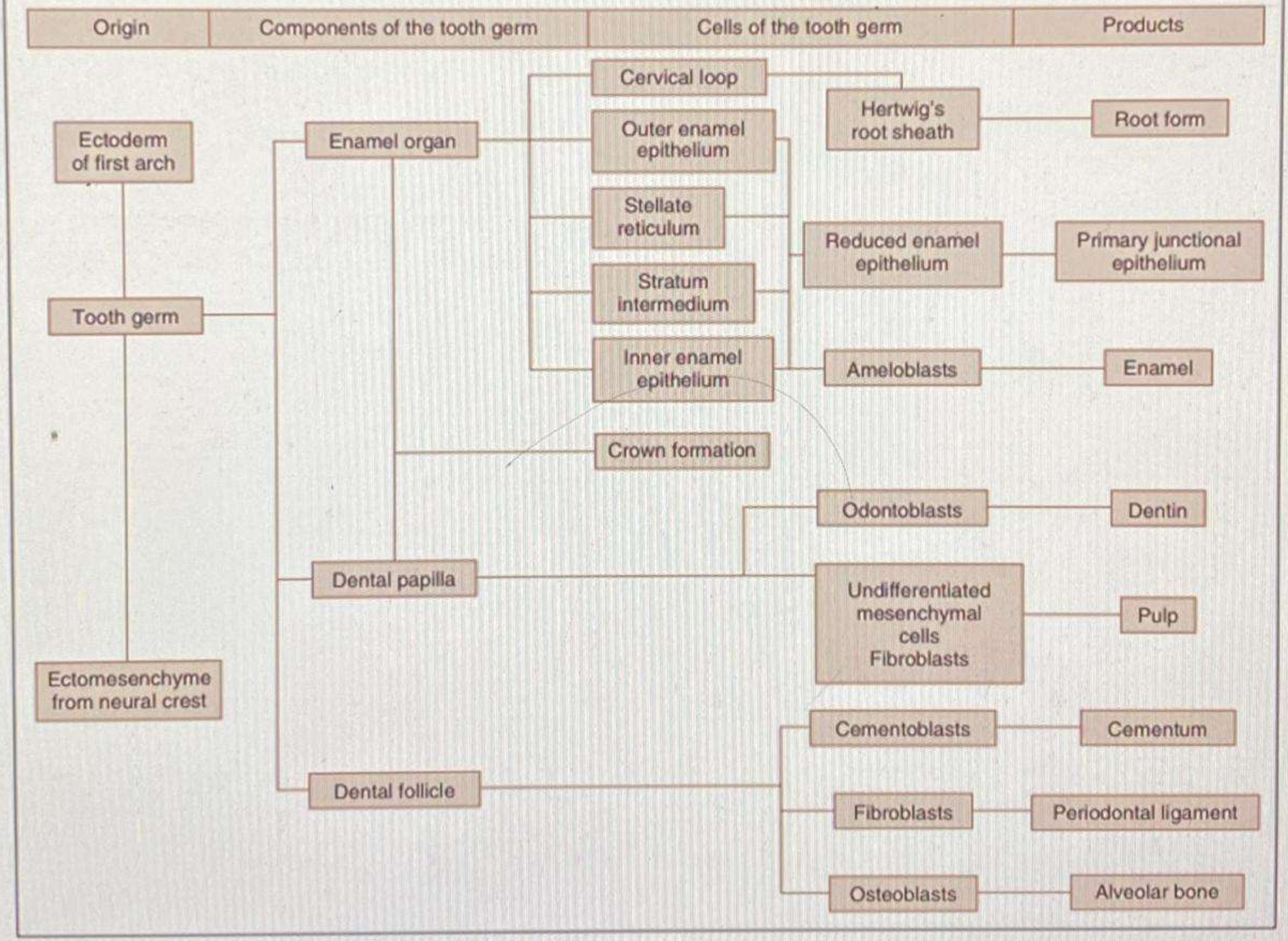
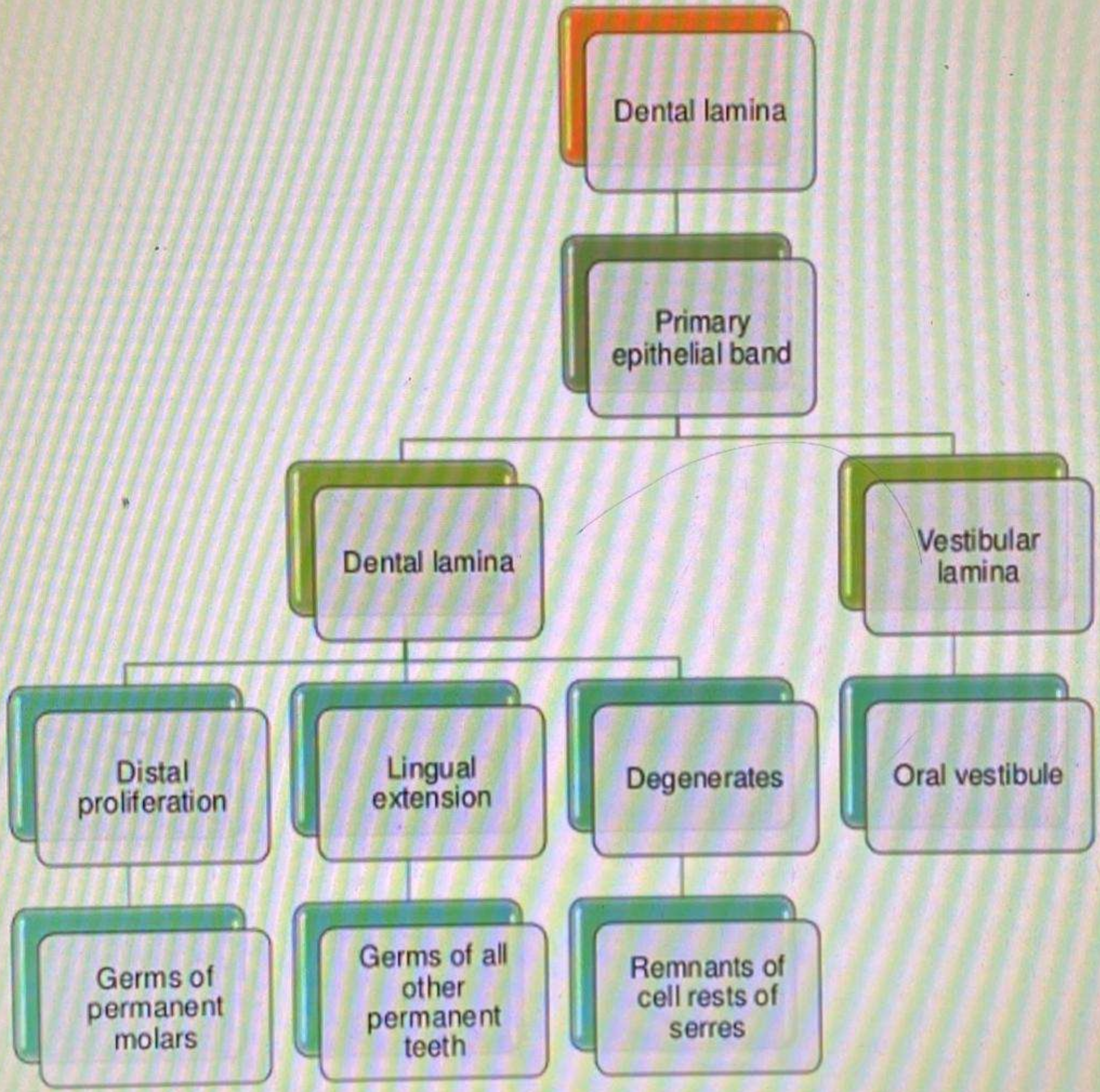


FIGURE 5-35 Summary of tooth formation.


Dental lamina
Primary epithelial band
Dental lamina
Vestibular lamina
Distal proliferation
Lingual extension
Degenerates
Oral vestibule
Germs of permanent molars
Germs of all other permanent teeth
Remnants of cell rests of serres

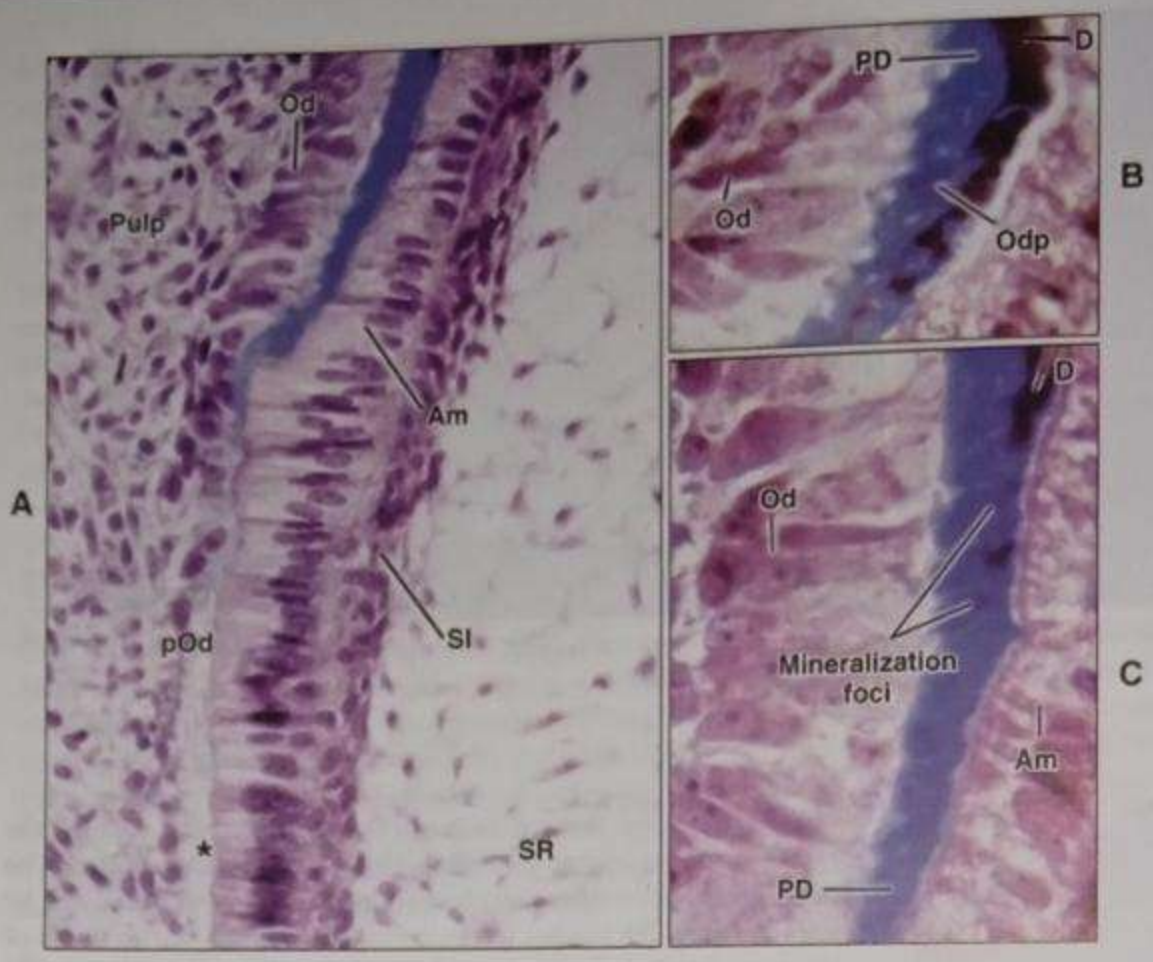


FIGURE 8-10 Changes in the dental papilla associated with initiation of dentin formation. **A,** An acellular zone *(*)* separates the undifferentiated cells of the dental papilla (preodontoblasts, *pOd*) from the differentiating inner enamel epithelium (ameloblasts, *Am*). Preodontoblasts develop into tall and polarized odontoblasts (*Od*) with the nucleus away from the matrix they deposit at the interface with ameloblasts. **B** and **C,** The matrix first accumulates as an unmineralized layer, predentin *(PD)*, which gradually mineralizes to form mantle dentin (*D*). *Odp*, Odontoblast process; *Sl*, stratum intermedium; *SR*, stellate reticulum.

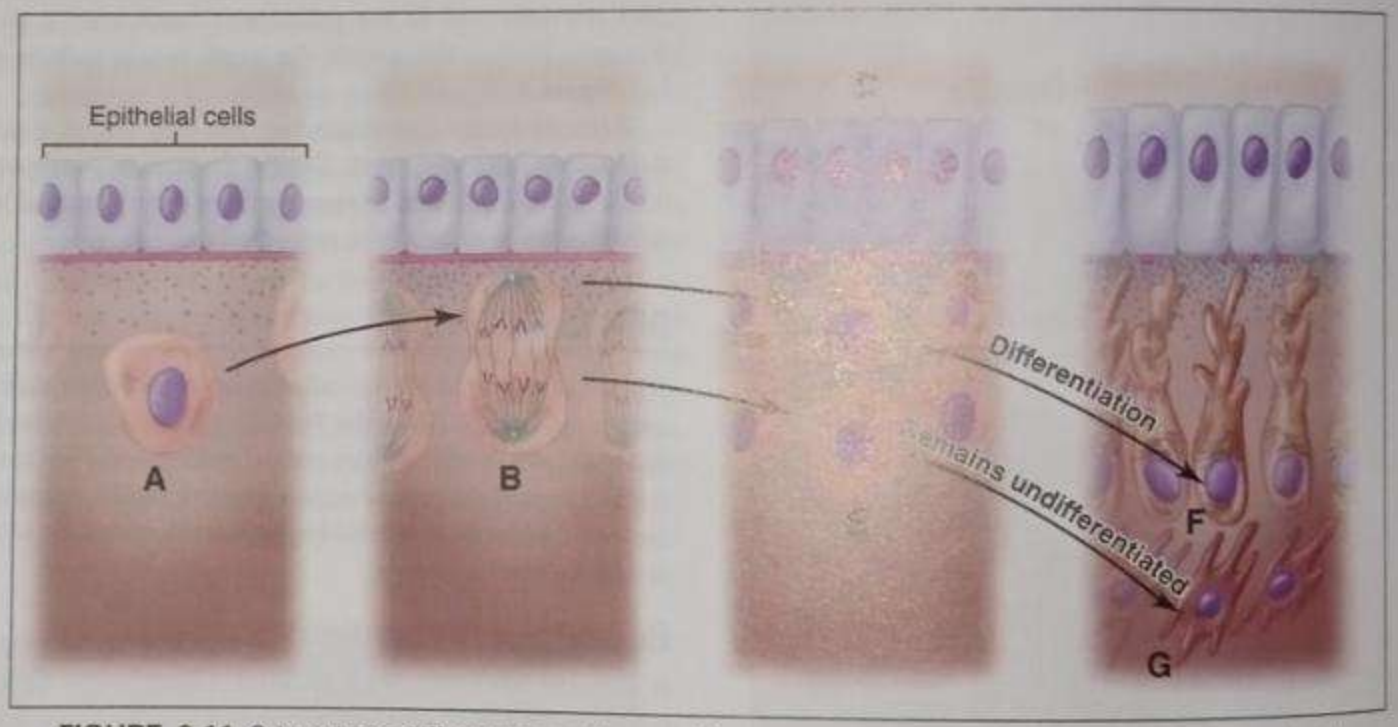


FIGURE 8-11 Odontoblast differentiation. The undifferentiated ectomesenchymal cell *(A)* of the dental papilla divides *(B)*, with its mitotic spindle perpendicular to the basal lamina *(pink line)*. A daughter cell *(C)*, influenced by the epithelial cells and molecules they produce *(D)*, differentiates into an odontoblast *(F)*. Another daughter cell *(E)*, not exposed to this epithelial influence, persists as a subodontoblast cell *(G)*. This cell has been exposed to all the determinants necessary for odontoblast formation except the last.

zone between the dental papilla and the inner enamel epithelium gradually is eliminated as the odontoblasts differentiate and increase in size and occupy this zone. These newly differentiated cells are characterized by being highly polarized, with their nuclei positioned away from the inner enamel epithelium.

Formation of Mantle Dentin

After the differentiation of odontoblasts, the next step in the production of dentin is formation of its organic matrix. The first sign of dentin formation is the appearance of distinct, large-diameter collagen fibrils (0.1 to 0.2 µm in diameter) called *von Korff's fibers* (Figures 8-12 to 8-15). These fibers consist of collagen type III associated, at least initially, with fibronectin. These fibers originate deep among the odontoblasts, extend toward the inner enamel epithelium, and fan out in the structureless ground substance immediately below the epithelium. As the odontoblasts continue to increase in size, they also produce smaller collagen type I fibrils that orient themselves parallel to the future dentinoenamel junction (Figure 8-16). In this way, a layer of mantle predentin appears.

(primary palate)

Nasal septum

Lateral palatine process

(Palatine shelf)



6 WEEKS

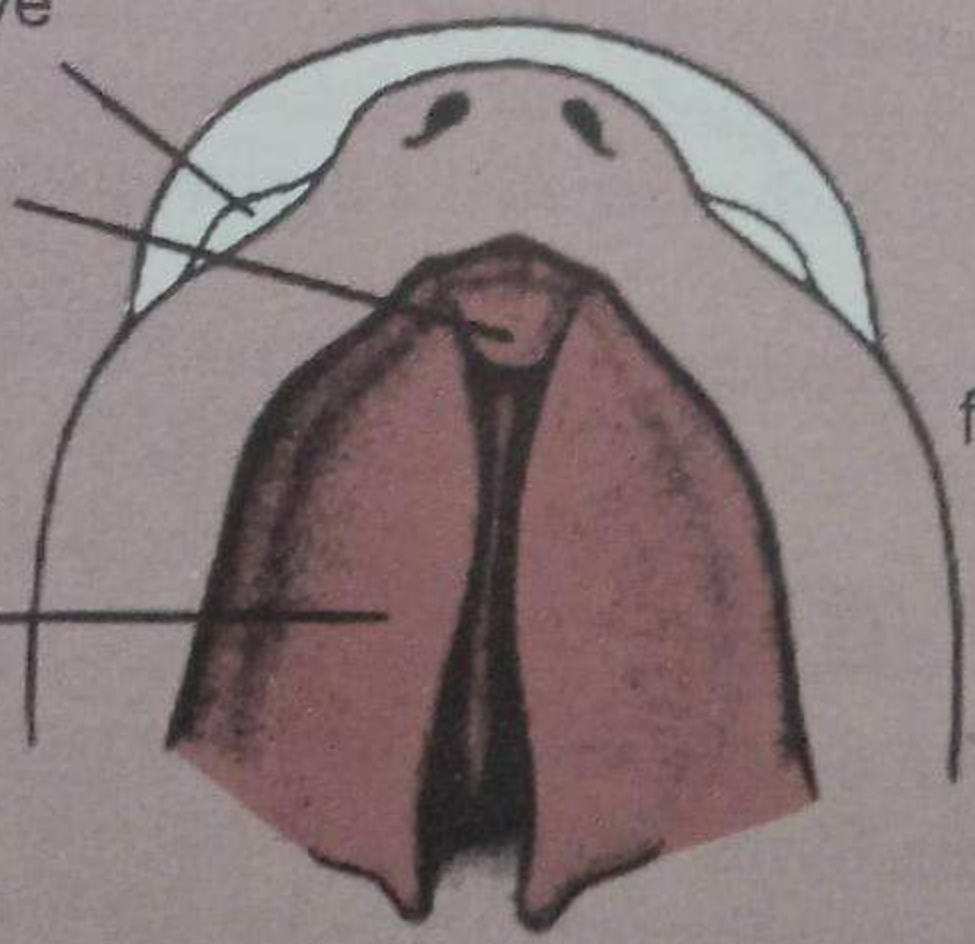
Incisive

Eye

primary palate

fused palate shelves

Palatine shelf



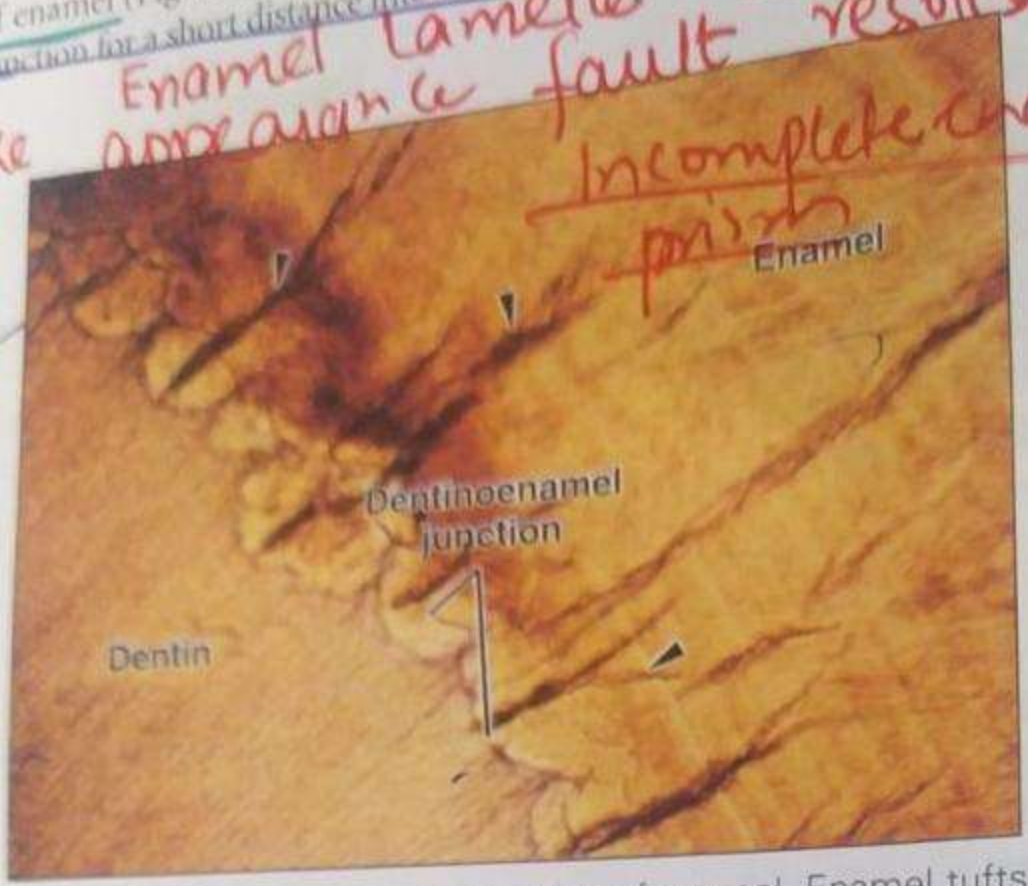
7 WEEKS

Enamel Tufts

Enamel tufts and lamellae may be
no known clinical significance. They are best seen
of enamel (Figure 7-57). Enamel tufts project from the dentino
junction for a short distance into the enamel. They appear to be branched

Enamel lamelle are sheet
ke appearance fault results in-

Incomplete enamel prism



IGURE 7-57 Transverse ground section of enamel. Enamel tufts are
e branched structures extending from the dentinoenamel junction into
e enamel (*arrowheads*). The junction is seen as a scalloped profile.

Dentinoe

The junctio
hard tissues
section (Figu
enamel form
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to form ena
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7-31). The
series of rid
the adheren
regard it is
dentin, wh
shape and
function.

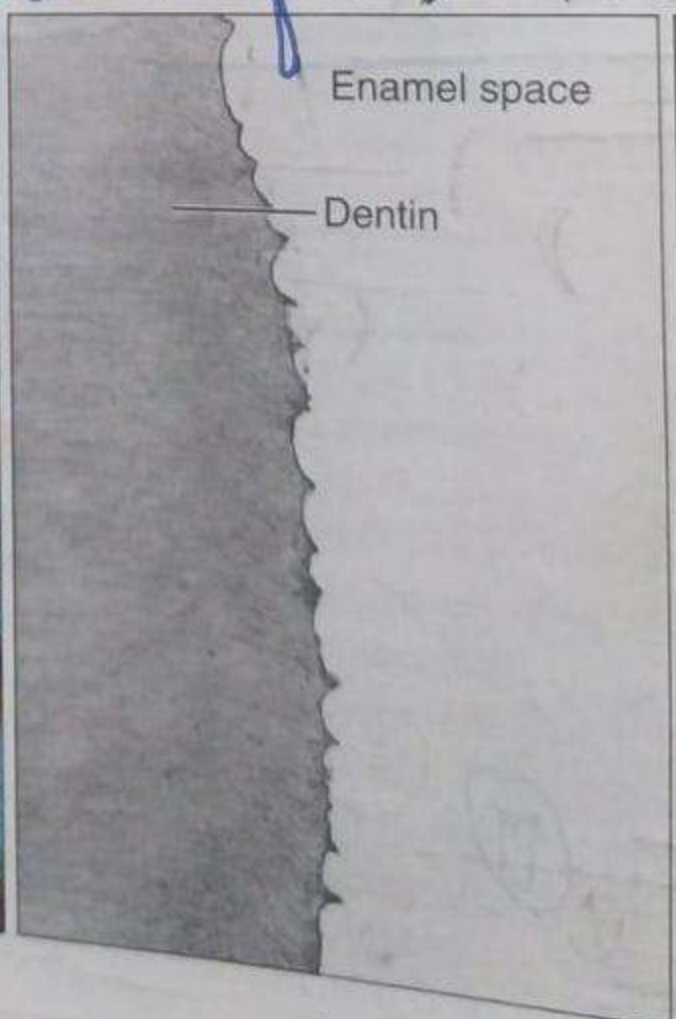
Enamel S

The surfac
of Retzius
surface of

Enamel lamellae are sheet-
faults as result of ⇒ incon



A



B

FIGURE 7-58 Dentinoenamel junction. **A,** Ground section
been lost. The scalloped nature of the junc
electron micrograph of a

- The upper canine teeth are called as the "eye teeth" as these have long roots which reach up to the medial angle of the eye. Infection of these roots may spread in the facial vein and even lead to thrombosis of the cavernous sinus.

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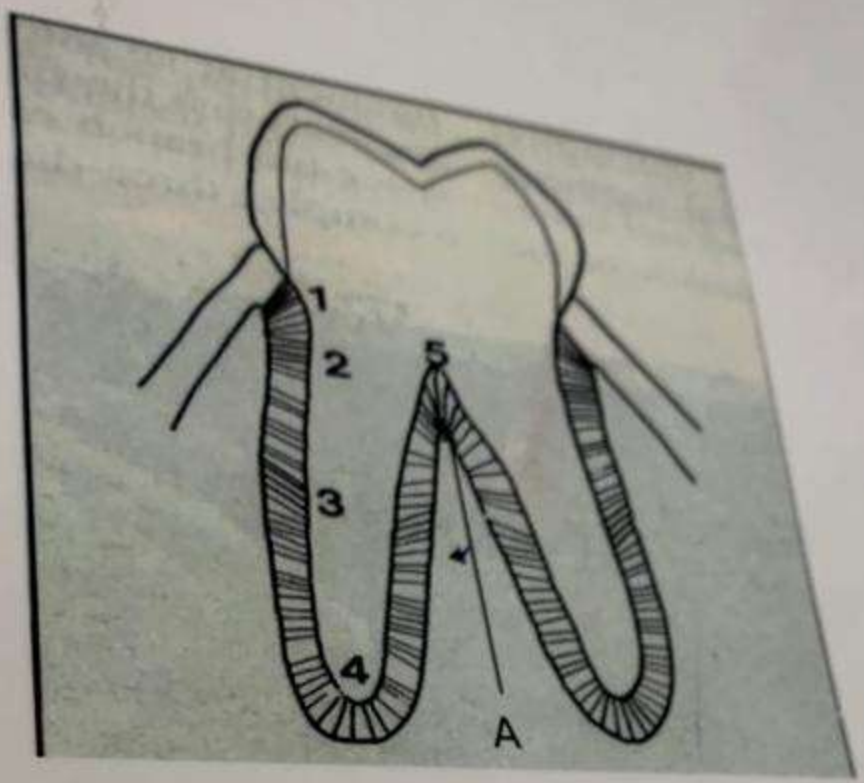


47 Identify the slide :
a) Gingival tissues
b) Epithelium
c) Connective tissue
d) Biological width

48 Identified structure contains:
a) Epithelium
b) Connective tissue
c) Epithelium + connective tissue
d) Gingival sulcus

Reply...

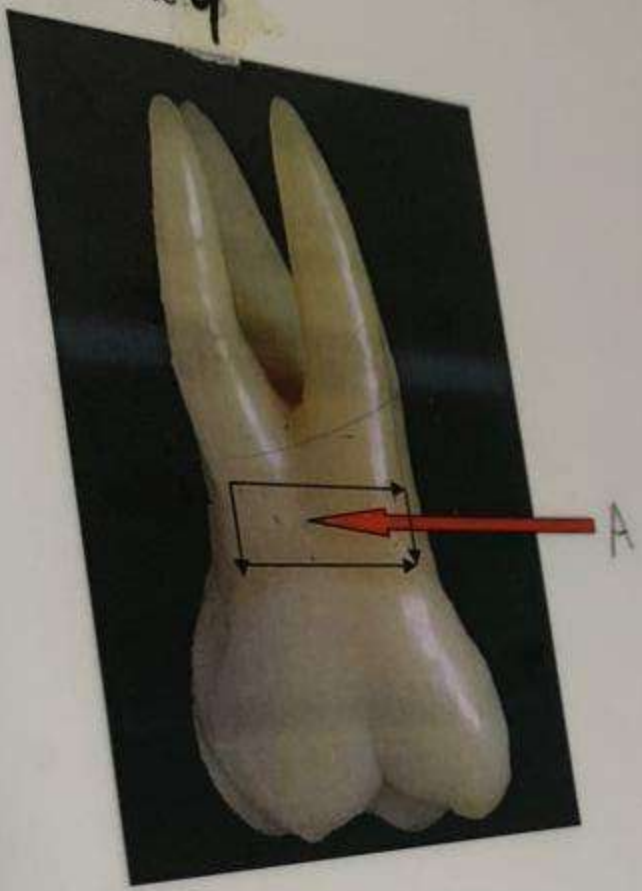
Station No 8



15. Identify the spot.
    a) The trans-septal ligament
    b) Alveolodental ligament
    c) Gingival fibers
    d) Principal fibers

16. Identify the spot A
    a) Alveolar crest group.
    b) Horizontal group.
    c) Inter-radicular group.
    d) Oblique group.
    e) Apical group.

Station No. 9



17. Identify marked spot A:
    a) Root trunk
    b) Root bifurcation
    c) Base of roots
    d) Cervical 1/3rd of root

18. Marked spot has length about:
    a. 3-4 mm
    b. 2-3 mm
    c. 4mm
    d. 1-3 mm